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RESEARCH ARTICLE

AI-Driven Innovation Using Multimodal and Personalized Adaptive Education for Students With Special Needs

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ABSTRACT This study provides an in-depth exploration of the use of multimodal techniques in developing adaptive learning systems designed for students with special needs using various neural network models: a Baseline Neural Network, Convolutional Neural Network, Attention Model, LSTM, GRU, and Transformer models. Adaptive learning systems play a major role in the customization of the educational process in such a way as to ensure the specific cognitive, behavioural and emotional needs of students with special educational needs are catered for. The models were trained using a sizeable amount of multimodal data that recorded factors such as academic achievement, behavioral interactions, and environmental factors. From our observations, it can be noted that transformer models and attention mechanisms allowed for higher performance in predicting and fulfilling the needs of students, and were more effective than simpler neural architectures. These findings suggest that more advanced and intelligent neural networks can improve how students with different learning abilities consume content and, consequently, provide better educational answers and solutions to the market. To identify the important features affecting student performance, various Explainable AI (XAI) techniques were utilized, including SHAP (Shapley Additive Explanations) values, logistic regression-based feature importance analysis, LIME (Local Interpretable Model-Agnostic Explanations), partial dependency plots, and whose global surrogate model was a Decision Tree Regressor. These XAI instruments allowed the model's decision to be explained and visualized so that teachers could understand the main beneficiaries of the improved educational results. Such XAI methods enable transparency in how models make decisions, thus allowing educators to understand the reasons for certain recommended learning paths.

INDEX TERMS Adaptive learning paths, students with special needs, neural networks, convolutional neural networks, attention mechanisms, long short-term memory, gated recurrent unit, transformer models, explainable AI.

I. INTRODUCTION

Artificial, Intelligence (AI) has changed in almost all fields and progressive solutions have been provided in terms of time, accuracy, and individuality [1]. Starting from diagnosing patients or identifying fraud in finances to market trends or predicting the spread of diseases, AI applications offer incredible computational resources to extract information that cannot be achieved manually [2]. The main driving forces of these developments are innovations in Machine Learning

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(ML) and Deep Learning (DL), whereby Smart Systems learn from patterns within large datasets, and then engage in tasks such as image recognition, natural language processing, and predictive analytics with unmatched accuracy [3]. All of these technologies allow systems to learn responses to specific contexts and change knowledge as it progresses, which is critical for environments requiring flexibility similar to that of humans.

Among the AI techniques, neural networks are crucial, with deep neural networks (DNN) at the core of adaptive applications. Neural networks are formed as layers of interconnecting nodes that simulate the neurons in the human brain. This architecture enables DNNs to detect and learn from different patterns within the data, such as language, images, or behaviors [4]. In adaptive learning, neural networks can incorporate many data inputs, such as learner responses, participation rates, and learning achievements, to produce specific student content and modify the learning processes. Progress in Convolutional Neural Network (CNN) and Recurrent Neural Network (RNN) structures coupled with new architectures such as transformers has extended the possibility of adept AI in managing and interpreting multimodal data, which are crucial in educational engineering [5].

However, the idea of adaptive learning still has a long way to go; hence, it faces many challenges, especially for disabled students. Some of the constraints of traditional learning systems include the inability to address different learning capabilities, learning models, and needs, owing to a variety of learning deficiencies, thereby reducing the reachability and efficiency of learning instruments [6]. The most challenging types of learners who might still require better and more personalized adaptive systems are those with cognitive, sensory, or physical impairments, because most standard adaptive environments are typically designed to handle only one and sometimes two modalities for input and output, but ignore how such a learner would best engage in a learning environment [7]. This gap shows that there is a strong need to develop learning models that incorporate more than one response style to teach, train, and inform students about diverse needs for sensory and interaction modes.

The absence of individuality in present-day AI-based adaptive interfaces amplifies the learning gap due to poor feedback and supports students with learning disabilities [8]. For instance, most existing AI models predominantly work with the text of images that are inaccessible to students with visual or auditory disabilities. Second, existing adaptive learning algorithms do not consider customized details because they provide generalized responses when, for instance, a student with multisensory learning impairment uses the system [9]. Thus, there is a gap between AI-enabling teaching tools and specialized learning needs that need to be corrected through the proposed multimodal adaptive approach, which can help provide greater access to learning for students with learning disabilities.

These problems have spurred the development of new and practically applicable adaptive learning strategies that incorporate multiple ML and DL models. This approach combines architectures such as CNNs for images, LSTMs for temporal information, and transformers for contextual computations into one system to process and produce learning in different formats. Multimodal adaptive learning systems can handle more data from students'sensory modalities and provide individual learning pathways that can accommodate disabled students [10].

The remainder of this paper is organized as follows: Section II discusses various studies related to AI in education, ML models used in adaptive learning systems, Neural Networks used for personalized learning, CNN used for visual learning, RNN for spatial data processing, transformer models used in adaptive learning, and DL models for multimodal adaptive learning systems. In Section III, various DL models are trained and evaluated in addition to Exploratory Data Analysis (EDA). Section IV discusses the use of the transformer model to generate adaptive learning paths. Section VI discusses the conclusions, limitations, and scope of future work.

II. BACKGROUND AND LITERATURE REVIEW

A. AI IN EDUCATION: BROAD OVERVIEW

AI integration in education has been a significant boost in creating better learning systems, customarily for curriculum and personalized learning. Studies such as [11] provide a primary background on deep learning architectures and how CNNs and RNNs process large datasets from which the model must learn for adaptation. One of the advantages of this study is the analysis of architectures for basic pattern recognition, based on the adaptation of learning machines. However, these studies often do not report practical examples of SN students, implying that generalizing the benefits of such technologies to this group of children is difficult.

CNNs are invaluable for processing visual learning data for content engagement and automated evaluations [12]. In engagement tracking, students' facial expressions during recorded or live lectures are analyzed using CNNs to assess whether they are interested or bored. If a student is disengaged from the content, the system can lower the content difficulty level or add motivating features in an attempt to regain student attention. In addition, automatic grading of handwritten answers can be provided through feedback using CNNs' recognition of digits, letters, and diagrams. CNNs further perform content recommendation by recognizing images in educational graphics so that the system can recommend additional materials based on the interaction of the student with resources.

On the other side, RNNs assist in learning sequence analysis through pattern recognition of a student's progress to predict future learning trajectories [13]. When an individualized learning path is constructed, RNNs, including LSTM and Gated Recurrent Unit (GRU) models, examine the order in which students perform activities, such as time on lessons and quizzes. This enables the system to anticipate the learner's needs in later stages and adapt the lesson plans proactively.

B. MACHINE LEARNING FOR ADAPTIVE LEARNING SYSTEMS

Adaptive systems also involve identifying performance and preferences for content delivery and feedback, which can be successfully handled with the help of machine learning models. Discussions of these systems by [14] highlighted the features of flexibility and how neural networks modify learning paths. Their study demonstrates that these models are flexible in various educational settings. Nevertheless, a limitation of these models is that they are highly dependent on large labelled datasets, which may not be easily found in the context of specialized education for learners with special needs. Thus, future studies that apply such models in these settings would require additional datasets.

C. NEURAL NETWORKS AND PERSONALIZED LEARNING

Neural networks are advantageous, particularly for applications in which each learner requires different input data from which they are capable of learning. Many studies have successfully applied neural networks in adaptive education because they can consider the varying rates and styles of student learning. For example, [15] discussed the learning of students using neural networks and their enhancing effects on the engagement and learning aspects of students with deficits. Something that remains unclear in this study is that despite its focus on the strengths of neural networks in terms of customization of learning, the approach could not satisfy the full needs of students with sensory disabilities since neural networks mostly work with either images or texts.

D. CONVOLUTIONAL NEURAL NETWORKS (CNNS) FOR VISUAL LEARNING

CNNs are best exploited for handling visual data, making them suitable for handling images and visual content data. This ability should be used in educational applications that offer a visual means of teaching, which is useful for particular learning disabilities. For example, as demonstrated in [16], CNNs have advantages when dealing with visual input and image classification for learning, and may be used as adaptive learning tools in education. However, CNNs do not model sequence or temporal data, which presents a problem for any application that needs to understand the language or behavior over time.

Because of their capability to extract hierarchical features, CNNs perform best with visual data. For instance, ResNet-50 is popular in image-based engagement detection, in which students' facial recognition expressions are evaluated to determine the levels of attention in adaptive learning environments [17]. Similarly, the VGG-16 was used to evaluate handwritten responses to enable automated scoring and feedback [18]. In aiding students with dyslexia, CNNs based image classification models help automate the grading and feedback processes by improving font readability through the alteration of font, spacing, and contrast. These examples demonstrate the importance of CNN architectures in improving adaptive learning using CNN's.

E. RECURRENT NEURAL NETWORKS (RNNS) AND SEQUENTIAL DATA PROCESSING

LSTM and GRU are highly efficient for analyzing language and time-dependent learning interactions, because RNNs are designed to work on sequential data. These networks perform well in settings where an adaptive learning system interacts with learners and the context must be interpreted over time. Goodfellow et al., state that these areas are perfectly suited for use by LSTMs and GRUs since they are sequence-dependent activities, like language processing and students' interaction patterns [14]. Nevertheless, a drawback of RNNs is their poor computational efficiency and vanishing gradient, which can hamper their performance in real-life educational applications if not optimized.

F. TRANSFORMERS IN ADAPTIVE LEARNING

These transformer architectures have undergone a shift to utilize contextual information in language-processing applications. The BERT and GPT models, among others, are transformers that can process the context of words and phrases and offer useful feedback and responses in flexible learning spaces. Vaswani et al. detailed the effectiveness of transformers in manipulating long-range dependencies of text data necessary for adaptation [19]. However, transformers are excellent in context-aware applications, and their inherent structure and training demands prove to be crucial barriers, despite their mightiness, especially in learning-deficient environments such as education.

Adaptive learning systems pose significant challenges for serving students with disabilities [20]. Conventional information technology in learning has developed with some presumptions of learning deficiencies. Hence, they cannot fully address the learning deficiencies of individual students [21]. Studies on adaptive systems have shown that most do not incorporate sensory or cognitive impairment, which eliminates students with such conditions who require an alternative way of interacting. Nevertheless, one shortcoming of the work of Edyburn et al. is that it provides a critically important analysis of these difficulties, but does not propose concrete technical solutions for coping with them; thus, further investigation of adaptive multimodal systems is required [22].

A type of bid data or multiple-modality adaptive learning model was proposed to solve the problems and shortcomings of single-mode learning systems. Owing to the combined use of text, images, and sounds in the learning process, these systems can provide students with a wider range of needs, especially those with disabilities, with materials that would best suit their needs. A related argument was proposed by Rose & Meyer, who pointed out that multichannel learning environments increase people's interest because information can be comprehended in several ways [23]. However, there

are some limitations associated with the proposed approach, such as the technical difficulty in facilitating different data types and ensuring consistency between them, which can be problems in the practical realization of the system and its application.

G. DEEP LEARNING IN MULTIMODAL ADAPTIVE LEARNING SYSTEMS

CNN and transformers have been successful in predicting the function of genes together with multimodal adaptive learning using several types of inputs. Schwartz et al. operationalized these models in their study of multimodal learning environments to enhance accessibility by supporting variations in visual, aural, or textual interfaces [24]. Strengths such as the deployment of multimodal methodologies in actual educational contexts are qualities that enrich this study. Nevertheless, drawbacks are in question, the model learning and training process, and the necessity to use a substantial number of labelled samples for desired accuracy regarding a great variety of inputs.

When considering the future of adaptive learning, these types of systems based on newly developed advanced neural network architectures can help change special needs education. The further development of the application of neural models is of great interest. For example, the application of CNNs to visual tasks and transformers for contextual learning resulted in a multifaceted adaptive configuration [25]. This study also focuses on the possibility of using multiple AI architectures simultaneously to meet extant learning needs. However, there are still barriers to overcome, such as high computational costs, ethical concerns regarding the use of learners' data, and the enhancement of teachers' capacity in the use of these tools in learning environments.

Table 1 provides an overview of state-of-the-art research on the positive outcomes and some drawbacks of using multimodal strategies in adaptive learning for learners with special needs.

III. MATERIAL AND METHODS

Figure 1 shows the complete process of providing adaptive learning paths for students with special needs. Each stage of the process is explained as follows.

A. DATASET OVERVIEW AND FEATURE DETAILS

The dataset can be accessed from https://figshare.com/s/3ae5b09234338c01263a. The features used in the dataset was extracted from publicly available datasets provided by the National Center for Education Statistics (NCES) through their website https://nces.ed.gov/programs/coe/indicator/cgg/students-with-disabilities. This study considered a dataset designed to model heterogeneous learning profiles of students with different disabilities. The dataset includes attributes that enable one to create a comprehensive picture of a student's particulars including age, intellect, and overall performance. A student ID was assigned to each student, which contained all the children's data separately, thus enabling the study to be clear and effective. The dataset encompasses school-aged children and adolescents, particularly those aged 5–18 years, and depicts various advances in and levels of educational achievement of learners.

Regarding the male-female ratio, the dataset comprises 'Male,' 'Female' and 'Non-Binary' which are broad categories providing a holistic attempt to establish adaptive learning requirements of all genders. It is important to note this demographic diversity when assessing the impact of adaptive learning on various groups. The participants in the dataset were classified according to the nature of their learning disabilities. These include dyslexia; ADHD; autistic spectrum disorder; and visual, hearing, and intellectual disabilities. These types were selected because of their significance in educational performance and their effect on students' learning activities.

The dataset consists of a variety of features that are believed to capture the essential characteristics of students' cognitive and academic performance. An essential feature is reading comprehension score, which defines how well students interpret and understand written words. This score is particularly relevant in examining the learning patterns of students with learning disabilities such as dyslexia. Math score is another important feature that examines the degree of quantitative reasoning possessed by learners, which is necessary for the attainment of set educational goals in most subjects.

The concept of memory performance is captured by a memory score that describes the extent to which a student can wait for some time and remember or retrieve learned material. This feature sheds light on some of the cognitive processes associated with successful learning, particularly among students who are likely to experience memory problems. The attention score in the dataset indicates the level of concentration exhibited by the students, which is a key component for students with attention-related disorders such as ADHD. The addition of a visual-spatial score demonstrates the ability of students to see information and turn it around, a skill that is crucial to learners with visual problems.

In addition to cognitive measurements, the dataset also captures student metrics regarding interactions with adaptive learning platforms, which provides a better understanding of student engagement and learning actions and behaviors. Time was measured in minutes and presented as an indicator of students' degree of interaction with the learning tool offered on the platform. This metric allows us to establish the correlation between use and performance outcomes, and makes it possible to determine the dynamics of learning engagement.

The completed assignment metric represents the number of learning tasks performed by students within the range of 0–10, which indicates the level of student effort and engagement in the learning process. This function serves as an empirical indicator of student persistence and work patterns in an adaptive learning environment. A teacher

TABLE 1. Literature review on multimodal approaches to adaptive learning for students with special needs.

Research Manuscript	Author(s) and Year	Model Used	Results	Limitations
Predicting behavioral changes in students with special education needs using multimodal learning analytics	Chan, Wong, & Yum (2023) [26]	Multimodal Learning Analytics	Applied Behavior Analysis Therapy predictions achieved 98% accuracy and 97% precision using multimodal data from environmental, psychological, and motion sensors.	Limited generalizability outside specific settings in Hong Kong and Singapore.
Multimodal Composing in Special Education: A Review of the Literature	Bruce et al. (2013) [27]	Literature Review	Emphasized the role of technology in supporting compositions for students with disabilities, especially digital videos.	Lack of empirical data on the direct impact of digital video composition.
Building pipelines for educational data using AI and multimodal analytics: A "grey-box" approach	Sharma, Papamitsiou, & Giannakos (2019) [28]	"Grey-box" Multimodal Pipeline	Demonstrated effective predictions of student engagement and effort using multimodal data streams, such as eye-tracking and facial expressions.	May have scalability issues when applied beyond specific adaptive learning contexts.
Predicting learners' effortful behavior in adaptive assessment using multimodal data	Sharma, Papamitsiou, Olsen, & Giannakos (2020) [29]	Hidden Markov Model (HMM)	The model accurately predicted students' effort levels using multimodal data, such as eye tracking and EEG, to guide feedback in real time.	Limited sample size; results may vary with student population.
Reducing Language Barriers in Science for Students with Special Educational Needs	Boyle, Rizzo, & Taylor (2020) [30]	Universal Design for Learning (UDL)	Multimodal tools, such as graphics and technology, are used to reduce language barriers, showing the potential to improve science education for students with disabilities.	Further studies are needed to determine their effectiveness in diverse science-learning settings.
Classifying student dialogue acts with multimodal learning analytics	Ezen-Can, Grafsgaard, Lester, & Boyer (2015) [31]	Unsupervised Machine Learning	Enhanced classification of student dialogue using multimodal cues such as posture and gestures enables a better understanding of natural language interactions.	Limited by reliance on posture and gesture data alone, without linguistic features.
Evaluation of an adaptive learning system based on Multimodal Affect Recognition	Standen et al. (2020) [32]	Multimodal Affect Recognition	The adaptive system increases engagement and reduces boredom by tailoring learning content based on students' emotional states.	No significant achievement gains were found, limited to short-term engagement effects.
Enhancing Student Vocabulary Proficiency through the Effective Implementation of Multimodal Approaches	Nadia (2023) [33]	Multimodal Approach in Language	Multimodal strategies such as games and visual aids significantly improved vocabulary proficiency among students with limited English proficiency.	Effects of the study on linguistic skills other than vocabulary were not measured.
Multimodal Strategies in Teaching Children with Autism: A Discourse Analysis	Wibowo et al. (2020) [34]	Multimodal Instructional Strategies	Teachers used a combination of verbal and nonverbal cues, such as gestures and facial expressions, to support the learning of students with autism.	Limited to a specific cultural context and small sample size.

feedback score was also added as a measure of performance quality, as this measure enhances the assessment of student improvement and underscores the role of external evaluation. This score is measured in terms of teachers' assessments, and provides a counterbalance when supported by quantitative measures.

As this dataset covers demographic, cognitive, and engagement-based characteristics, it is useful for analyzing various types of approaches applied to adaptive learning for students with special needs. Thus, the integration of a vast number of features makes it possible to provide strong evidence of how adaptive learning systems can address

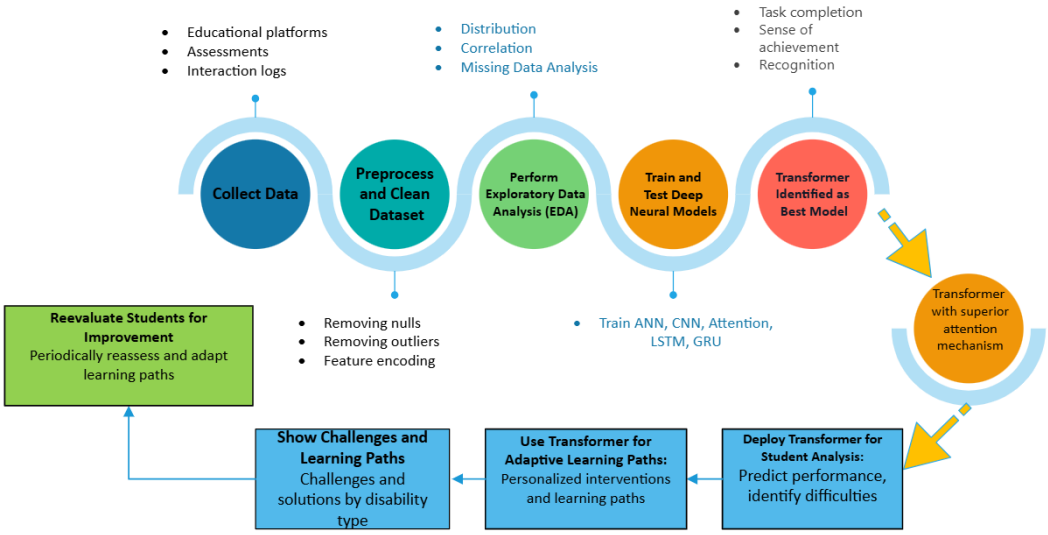


FIGURE 1. End-to-End process for adaptive learning for students with special needs.

various learning issues, and therefore advocate differentiated educational approaches that may fit learners' profiles.

B. DATASET PREPROCESSING

Ensuring the quality and reliability of the collected dataset prior to model training was very important, and the data preprocessing phase aided this. First, unnecessary columns, which included Unnamed: 0 and student_id, were removed to eliminate unwanted features. Missing values and duplicate records were checked and a thorough examination confirmed the completeness of the dataset, ensuring data integrity.

Categorical features were encoded in a systematic manner to allow compatibility with the machine learning models. For example, binary categorical features, such as gender, were label encoded, whereas multi-class categorical features, such as disability_type, were converted using one-hot encoding. To reduce scale discrepancies and standardize the numerical features, Minmax scaling was employed, ensuring that all features were within a comparable range.

Finally, the dependent variable's distribution was retained, and an efficient model evaluation was performed after performing an 80-20 train-test split. The data quality, consistency, and fair distribution obtained through these processing steps form a stiff foundation for subsequent actions.

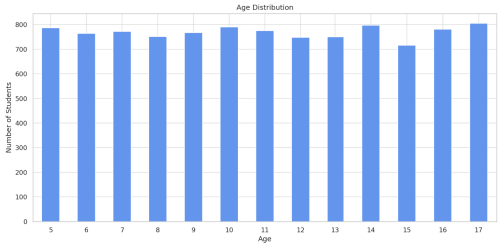


FIGURE 2. Bar chart depicting the distribution of students' ages in the dataset, ranging from 5 to 18 years.

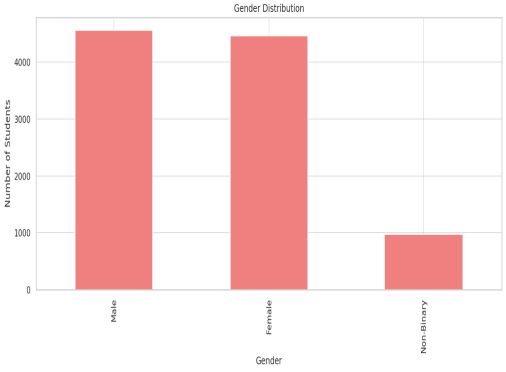


FIGURE 3. Bar chart showing the gender distribution of students. The dataset includes 45% 'Male' and 45% 'Female' students, with a 10% representation of 'Non-Binary' students.

C. EXPLORATORY DATA ANALYSIS

The age distribution graph in figure 2 indicates the ages of the students within the dataset with an age range of 5–18 years. This highlights the fact that the sample caters to young students in early childhood and adolescence, making it possible to analyze learning performance at two or more different developmental stages.

Based on the information in the gender composition bar graph the dataset achieved fair proportions of respondents who were 'Male' and 'Female' students at 45 percent each and 'Non-Binary' respondents at 10 percent as shown in figure 3. This distribution makes it possible to analyze the influence of gender on adaptive learning outcomes from a wider perspective.

The disability-type analysis shown in figure 4 indicates the different frequencies of the learning problems available within the dataset. Dyslexia, ADHD and Autism Spectrum Disorder were estimated to account for approximately 20%

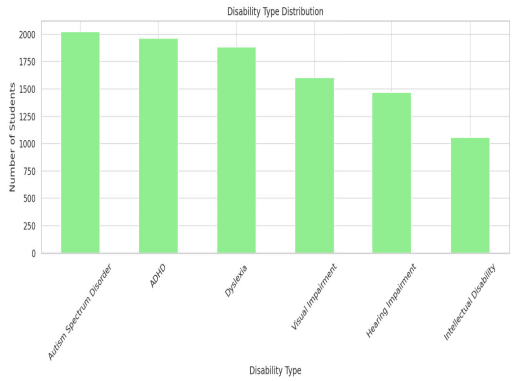


FIGURE 4. Bar chart representing the distribution of different types of disabilities within the dataset. The most prevalent disabilities include 'Dyslexia,' 'ADHD,' and 'Autism Spectrum Disorder (each approximately 20%), while 'Visual Impairment' and 'Hearing Impairment' each constitute 15%. 'Intellectual Disability' accounts for 10% of the sample, providing a comprehensive view of diverse learning needs.

of the representative sample, whereas visual and hearing impairments represented 15% each, and Intellectual Disability accounted for 10%. These proportions allowed for a comparative study of the different learning needs.

The histograms that portray reading comprehension, math, and memory scores outline the distribution patterns of the three metrics, as shown in figure 5. Most scores fell within a small range of average values. These metrics help understand how students perform cognitive tasks that are important for adaptive learning systems.

The distribution plots regarding the time spent on the platform and the number of assignments completed outline the way students engage with the material 6. The time spent was between 10 and 120 min, suggesting that there were different levels of time engagement, while the number of assignments ranged from 0 to 10, indicating the level of effort exerted by the students.

However, the bimodal histogram in Figure 7 shows the distribution of teacher feedback scores per student, indicating that most of the scores were high, and that many students received fair assessments. This feedback helps reconcile subjective estimates with objective measures of performance.

The bar chart for the final performance in figure 8 shows that 70% of the learners passed, and 30% of the learners failed. However, this distribution underlines the achievements more broadly, where a balanced view of the model training is evoked without skewness extremes.

D. EXPERIMENTS: BASELINE NEURAL NETWORK MODEL FOR ADAPTIVE LEARNING

A deep feedforward network was constructed and used as a baseline model to predict the student performance. Up to a certain point, the architecture includes an input layer, three hidden layers, and an output layer, each of which performs a certain role in terms of structuring and modelling the nonlinear relationships between the variables and the dependent variable, resulting in the construction of these layers. This network's input layer consists of points that cover student demographics, cognitive abilities, and level of engagement, thereby ensuring a wide span of student attributes.

A feedforward neural network with three hidden layers (128, 256, and 128) used the ReLU activation function. The model was trained for 200 epochs with early stopping based on validation loss. Overfitting was mitigated with dropout (0.3) and the Adam optimizer was used with a learning rate of 0.001. A grid search was implemented on the batch size, unit stride, and learning rate with values of (16,32), (0.01, 0.001), and (64, 128, 256), respectively. The best configuration was found for a batch size of 32, learning rate of 0.001, and 256 hidden units in the middle layer.

In addition, to help increase the level of abstraction over the features, hidden layers were made to decrease in the order of dimensions. The first hidden layer contained 64 neurons and used a Rectified Linear Unit (ReLU), which is a nonlinear activation to output layers used to model complex functions. After this layer, a dropout layer was applied at a rate of 0.3 while training 30% of the neurons to be dropped randomly. This step was performed to avoid overfitting the network, as it ensured that the network was not overly reliant on certain routes during its operation. The second hidden layer comprised 32 needs that were also activated by the ReLU, where another dropout layer was placed at 0.3 to further advance the cause of the reduction in overfitting. Finally, in the third hidden layer, 16 neurons were placed, where more abstraction of the learned information occurred, and the feature space representation was further simplified.

Two neurons in the output layer have a softmax activation function that provides an output that can be interpreted as a probability for binary classification. This design ensures that the model predicts the probability of classifying two classes–passing or failing–making decisions more straightforward. The fundamental principle of this neural network is to take the input vectors and transform them through fully connected layers constructed in a manner that takes a linear combination and, most of the time, applies a nonlinear function. The hidden layers use the ReLU activation function because they enable the model to learn about the nonlinear relationships between the features of the reading methodology, mathematics scores, attention, and other engagement metrics. Internal neural network dropout layers are also highly important for generalization because a few neurons are randomly inhibited as they are switched off, and many neurons do not focus too specifically on the training data in which the model is being trained. The last layer uses softmax activation; it provides class probability distributions, which sum to one, making it easier to interpret the output of the model for classification. The presented probability-based output provides accurate predictions and helps in the calculation of performance indicators such as accuracy.

The Adam optimizer was preferred because it has an adaptive learning rate that changes based on the first

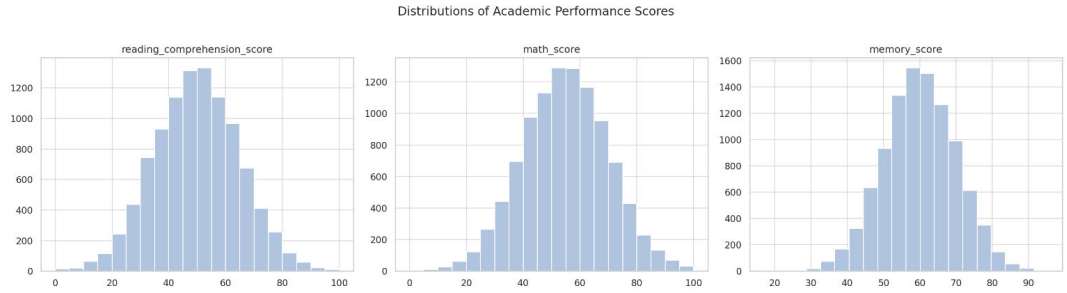


FIGURE 5. Distribution of academic performance scores for reading comprehension, math, and memory. The scores are centred around their respective means, with some variation, providing insight into students' cognitive abilities.

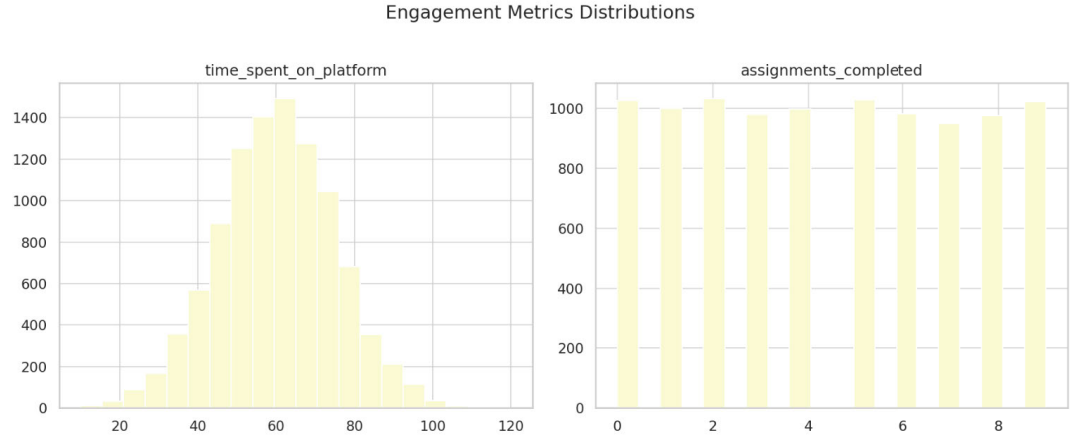


FIGURE 6. Histograms illustrating the distributions of engagement metrics, specifically the time spent on the platform and the number of assignments completed by students. The time spent ranges from 10 to 120 minutes, showing different engagement levels, while the assignments completed vary between 0 and 10, reflecting the varying degrees of effort and commitment among students.

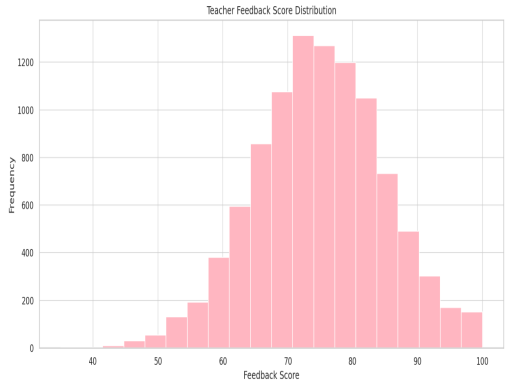


FIGURE 7. Histogram representing the distribution of teacher feedback scores, with most scores clustered around a high average. This distribution indicates that many students receive positive qualitative evaluations, which can be correlated with other performance metrics for deeper insights into student progress.

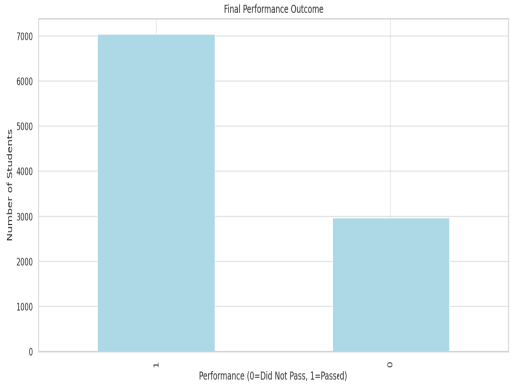


FIGURE 8. Bar chart showing the distribution of final performance outcomes. Approximately 70% of students passed, while 30% did not, indicating a moderately successful learning environment.

and second moments of the gradient. This enables quick convergence and better sparsity of gradients, which is suitable in this case because the datasets are high dimensional. Model training utilizes the categorical_ cross-entropy loss function that gauges the predicted class probabilities of the expected distribution. This loss function is appropriate for multiclass or binary classifications with one-hot-encoded target variables. Accuracy is the primary metric for measuring how well the training has been performed, which helps track the correct predictions of the model through training.

In this case, each epoch refers to a complete forward and backward pass over the training set. Thus, the network was

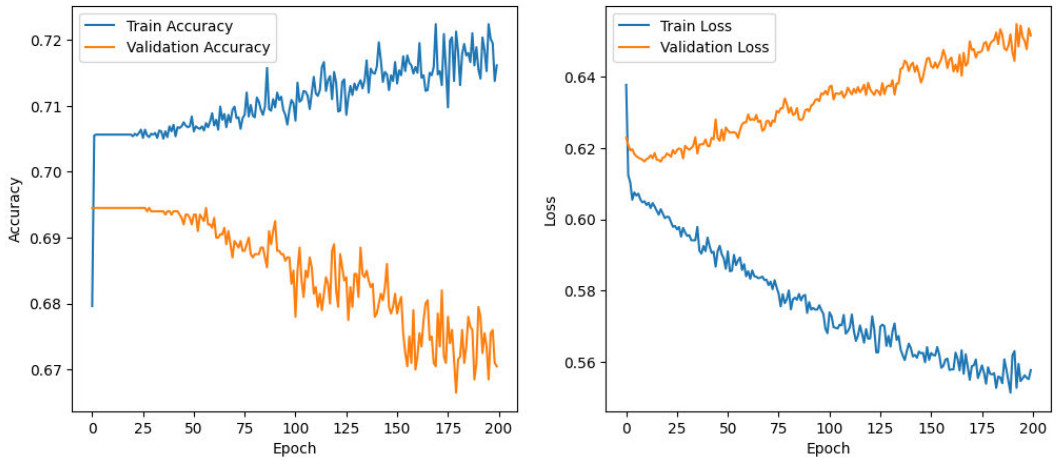


FIGURE 9. Training-validation accuracy and training-validation loss of the baseline fee-forward neural network.

TABLE 2. Baseline model evaluation metrics.

	Precision	Recall	F1-Score	Support
Class 0	0.21	0.03	0.05	611
Class 1	0.69	0.95	0.80	1389
Accuracy		0.67 (2000 samples)		
Macro Avg	0.45	0.49	0.42	2000
Weighted Avg	0.54	0.67	0.57	2000

trained for 200 epochs. In this case, a batch of 32 is considered appropriate because it represents an acceptable compromise between computational speed and consistency in the gradient updates. Such selection aims to ensure that the model receives as many examples in a batch as possible to learn effectively, while accommodating the limits of the system's memory.

Throughout the training period, the dataset was partitioned into training and validation sets such that the generalization ability of the model could be measured periodically during training. The batch of data is processed by the model, predictions for the input are produced, and the produced predictions are compared with true labels to generate losses using categorical_crossentropy. The weights were further modified using the backpropagation algorithm to reduce the loss depending on the gradient and hyperparameters using the Adam optimizer. Upon completion of the training phase, the trained model was tested on a test set that was not used in the training phase, to avoid introducing any bias in the evaluation of the model. This sequence of events is called the testing phase, and it assesses the model's ability to generalize new data. This is useful because it provides an insight into how an AI model can be utilized in a real-world scenario. To evaluate the predictions, the success metric used was quantitative accuracy, with other measures such as recall and precision utilized as the situation demands. The usefulness of this baseline neural network model provides an understanding of how multimodal educational data can be processed by using basic feedforward architectures. This model is also important because the results of this research set a benchmark in the search for more efficient models, such as recurrent neural networks and transformer architectures, which could be more efficient in capturing the sequential dependencies or interactions of multiple features. Table 2 lists the relevant evaluation metrics for the baseline model to predict students' performance outcomes. These metrics were defined as precision, recall, F1, and support for both classes (Classes 0 and 1).

Class 0 (students predicted not to pass): The model yields 0.21 precision, but the recall was only 0.03, leading to an F1 score of 0.05. This means that even though some correct decisions are made with respect to Class 0, the model does not perform well with respect to sensitivity for this class, and performs many false omissions or errors.

Class 1 (students predicted to pass): The metrics for this class differed significantly in terms of performance. The precision was 0.69, with a recall of 0.95, resulting in an F1 score of 0.80. The performance shows that this class has a strong passing ratio and the model can pass a substantial number of correctly passing students. 0.67 stands as the overall accuracy of the model, showing the general effectiveness for a dataset containing 2000 samples. For the macro average, the results returned (0.45 precision, 0.49 recall and 0.42 for F1 score) are reflective of the model performance for both classes without the bias of differential weights. The weighted average scores often changed owing to skew within the dataset, with a precision of 0.54, recall of 0.67, and F1 score of 0.57, indicating better performance in the large Class 1 group. Figure 9 shows the training and validation accuracies and loss of the baseline model.

E. TRANSFORMER MODEL

Next, a Transformer-based architecture was used to predict learners' performance. A custom Transformer Encoder Block implemented in TensorFlow is employed. This encoder block

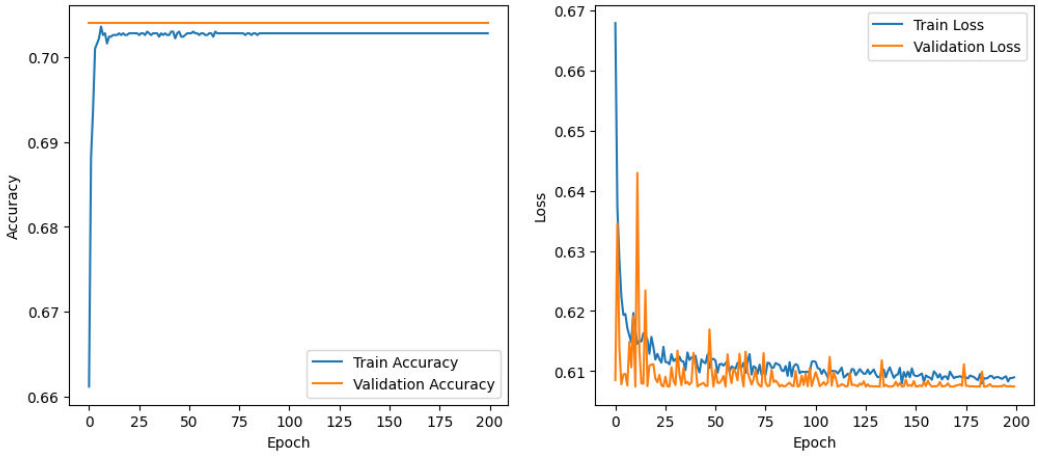


FIGURE 10. Training-validation accuracy and training-validation loss of the transformer model.

TABLE 3. Transformer model evaluation metrics.

Class	Precision	Recall	F1-Score	Support
0	0.00	0.00	0.00	611
1	0.69	1.00	0.82	1389
Accuracy	0.69			
Macro Avg	0.35	0.50	0.41	2000
Weighted Avg	0.48	0.69	0.57	2000

emphasizes certain relationships among input features using a multihead attention mechanism. This mechanism allows for the simultaneous consideration of several representation subspaces. The encoder is composed of MultiHeadAttention, which enables the parallel computation of attention, and two fully connected layers are arranged as a feed-forward neural network. A Layer Normalization (LayerNormalization) mechanism was also included because it guarantees the numerical stability of the model and improves the training efficiency by normalizing the inputs of each layer. Additionally, dropout layers were added to minimize overfitting when the model was trained.

A 6-layer transformer encoder with multihead self-attention (eight heads each) and feedforward layers of 512 dimensions was used. Positional encoding was applied to maintain sequence order. Pre-training on a broad multimodal dataset, prior to fine-tuning, was performed on the adaptive learning dataset. The optimizer was set to AdamW with a learning rate scheduler (10% of the steps were warmed and subsequently decayed). A batch size of 16 along with 200 epochs of training was implemented. Hyperparameter Tuning: A random search was performed over learning rates (5e-05, 3e-05, 1e-04), batch sizes (8, 16, 32), and number of attention heads (4,8,12). Optimal results were achieved using eight attention heads, a learning rate of 3e-05, and a batch size of 16.

The attention process comprises two serial operations: Transformer Encoder Block attention, and feature transformation. First, the input passes through the multiheadattention layer and reaches an attention-weighted output, which focuses on the most significant portions in the context of the input sequence. This output is then summed up with the input and undergoes the layer normalization process, which is a standard procedure known as Addition and Normalization. In the second stage, the output after normalization is fed into a feedforward network (dense layers) to enhance the representation of the features. The final step involves transforming the output features by applying the ''Add & Norm'' step to the attention output. Dropouts were applied at each stage to generalize the model.

This Transformer Encoder Block acts as a simple building block when stacking transformers for time-series data, which includes the time spent on activities, the number of answers provided, and explanations of previous questions. These steps allow the model to use encoded pieces of information to determine whether a learner is able to answer the next questions correctly and assist in pattern recognition during the learning process. Focusing on multihead attention and feedforward networks, the architecture defines both time dependencies and feature relationships, which are important for evaluating learner performance.

The essential component of the model is a custom TransformerEncoderBlock that uses multihead attention (MultiHeadAttention) to attend to relevant portions of the input sequence, and a feed-forward network to enhance feature representation. The block incorporates dropout layers to counterbalance overfitting and Layer Normalization to enhance training stability. This encoder block accepts the input sequentially and forms a part of an entire transformer that uses time-series data (input layer). The output of the encoder block was squeezed (flattened) and configured to work with a dense layer (dense) using a softmax function to predict the probabilities of a given outcome, such as the appropriateness of the learners' answers. A categorical cross-entropy loss function with the Adam optimizer was used to train the model, and its performance was evaluated in terms of accuracy metrics over several epochs, thereby demonstrating

TABLE 4. Deep neural network model performance summary.

Model	Accuracy	Precision	Recall	F1 Score
Baseline NN	0.6705	0.690302	0.953204	0.800726
CNN	0.6210	0.701340	0.791217	0.743572
Attention	0.5920	0.703336	0.713463	0.708363
LSTM	0.5815	0.692201	0.715623	0.703717
GRU	0.5720	0.686233	0.706983	0.696454
Transformer	0.6945	0.694500	1.000000	0.819711

the capability of the model to analyze sequential educational data.

Table 3 provides an insightful analysis of the transformer model performance in the test dataset. There are subclass metrics for precision-recall and F1-score for both parts: Class 0, the participants who provided incorrect answers; and Class 1, the participants who provided correct answers. Class 0 metrics indicated that the performance in the identification of other incorrect metrics for this classification achieved a zero score. However, Class 1, which clustered the correct answers, had a very good recall (1.00) and reasonable F1 score (0.82), which confirms that the model will be able to consistently find correct answers.

In this case, the model exhibited a moderate performance accuracy of 69%. The macro average scores (precision: 0.35, recall: 0.50, F1-score: 0.41) measure are presented, which show the average across all classes and, hence, the gap in inequality between class performances. Weighted averages (precision: 0.48, recall: 0.69, F1-score: 0.57) measures show the average across class values, and therefore show a helpful measure of accuracy when determining the overall average accuracy of the model. This evaluation indicates that the model has high efficiency in the prediction of correct answers, whereas soft spots exist when defining false answers. Figure 10 shows the training and validation accuracies and losses of the transformer model.

Table 4 summarizes the evaluation of various deep neural network models, including the Baseline Neural Network (NN), Convolutional Neural Network (CNN), attention-based model, long short-term memory (LSTM), Gated Recurrent Unit (GRU), and transformer models, for four parameters: accuracy, precision, recall, and F1 score. Of these models, the transformer model scored the highest accuracy at 69.45% while achieving an ideal recall value (1.00), suggesting that this model is good at retrieving all relevant instances correctly. The Transformer model also possessed the highest F1 score of 0.819711, making it ideal for moderation between precision and recall.

The Baseline NN model performed accurately, with accuracy values of 67.05% and very high recall values of 0.953204, which are crucial for its F1 Score, with values of 0.800726. Similarly, the CNN model exhibited moderate performance with an accuracy of 62.10% and an average F1 score of 0.743572. In general, the Attention, LSTM, and GRU models exhibited lower accuracies, with attention performing marginally better than the LSTM and GRU models. These findings highlight the advantages of the transformer model in terms of learner performance prediction and support its application in this area compared with other NN architectures.

TABLE 5. CNN model evaluation metrics.

	Precision	Recall	F1-Score	Support
Class 0	0.33	0.23	0.27	611
Class 1	0.70	0.79	0.74	1389
Accuracy		0.62 (2000 samples)		
Macro Avg	0.52	0.51	0.51	2000
Weighted Avg	0.59	0.62	0.60	2000

F. CONVOLUTIONAL NEURAL NETWORK TRAINING AND TESTING

A Convolutional Neural Network (CNN) model suitable for pattern learning was developed, trained, and tested to predict the student performance. The model consists of three layers, where the initial layers have one convolutional one-dimensional Conv1D with 32 filters and a kernel size of three pieces, with ReLU as the activation function. This layer extracts the local patterns and relevant feature interactions from the input data. Third, a dropout layer with a fraction of 0.3 is introduced as an additional regularization technique for the model to overcome potential overfitting by switching off a proportion of some of its units at random during the training of the network. Second, a Conv1D layer, which has 16 filters but the same activation function, is used to deepen feature extraction. More abstractions of the data characteristics were captured. Third, the output of the convolutional layers is fed into a flattened layer that outputs the feature graph into a one-dimensional array. This is such that the dense layer can receive them immediately after the flattened layer, which

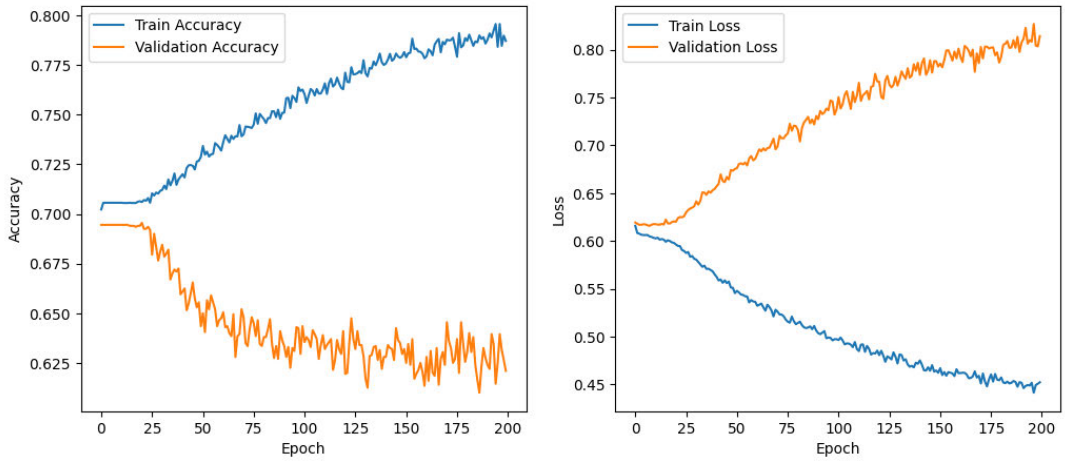


FIGURE 11. Training-validation accuracy and training-validation loss of the CNN.

further expands the output of the flattened filter to 16 neurons. The last layer is the softmax layer for probabilities obtained from the two neurons of the dense layer, which lasts in the entire network and is built to provide the linear rate and/or binary outcomes of student performance prediction in this case.

The CNN model possesses four convolutional layers: batch normalization and max-pooling. Fully connected layers were added after the classification. The model considers the cross-entropy loss and is trained with an Adam optimizer on an lr of 0.0001. Trained for 200 epochs with early stopping if the validation loss did not improve in five epochs. Grid search on the number of filters (3,4,5), filter sizes of 32,64,128 and learning rates of 0.01,0.001,0.0001. The optimized model had four layers and 64 filters at a learning rate of 0.0001.

The training procedure of the CNN model is such that it is compiled using the Adam optimizer and categorical cross-entropy loss, because both are good for classification. The model was iteratively trained for over 200 epochs with a batch size of 32, and a backpropagation technique was used to update the weights and minimize the categorical cross-entropy loss function. During training, validation data are available to assess the performance of the model, and the trend towards overfitting or underfitting can be observed early. The increased input shape works well, such that the data are reshaped to fit the Conv1D layer without losing the sequential order of the data. Finally, the model was tested on a test dataset to evaluate its performance in terms of generalization to unseen data. Measures, such as network accuracy, were used to ensure that the network accurately captured the temporal structure of the dataset.

Table 5 summarizes the evaluation metrics of the CNN model developed for forecasting the student outcomes. Class 0 (students who were not expected to pass the course) had a precision value of 0.33, recall value of 0.23, and F1-score value of 0.27, which shows that the vast majority of students who were expected not to pass the course could fail with no significant likelihood that they would fail. Class 1 (students predicted to pass) appears to be better off, with precision-recall and F1-scores higher than those of the previous class, at respective values of 0.70, 0.79, and 0.74, indicating that several students have a strong chance of passing the course level.

TABLE 6. Attention model evaluation metrics.

	Precision	Recall	F1-Score	Support
Class 0	0.33	0.32	0.32	611
Class 1	0.70	0.71	0.71	1389
Accuracy		0.59 (2000 samples)		
Macro Avg	0.51	0.51	0.51	2000
Weighted Avg	0.59	0.59	0.59	2000

In general, the accuracy rate of the model is 0.62, which is a mediocre performance on a dataset with 2000 samples. In addition, the macro average precision, recall, and F1-score yielded values of 0.52, 0.51, and 0.51, respectively, indicating average performance for both classes. The weighted average considers the proportion of instances of the classes and gives the values of 0.59 for precision, 0.62 for recall and 0.60 for the F1 score, which implies that the model is relatively good at predicting the majority class, while the minority class (Class 0) can be better managed. Figure 11 shows the training and validation accuracies and loss of the CNN model.

G. ATTENTION MODEL

In this paper, we propose an attention-based model that improves the network approach by focusing on important input features. The architecture begins with an input layer that is used to resize the input data for compatibility with the attention mechanism. Two fully connected layers with 64 neurons activated by Rectified Linear Units (ReLUs) were then implemented to obtain a multiple-feature space of the input. These layers facilitate feature extraction before

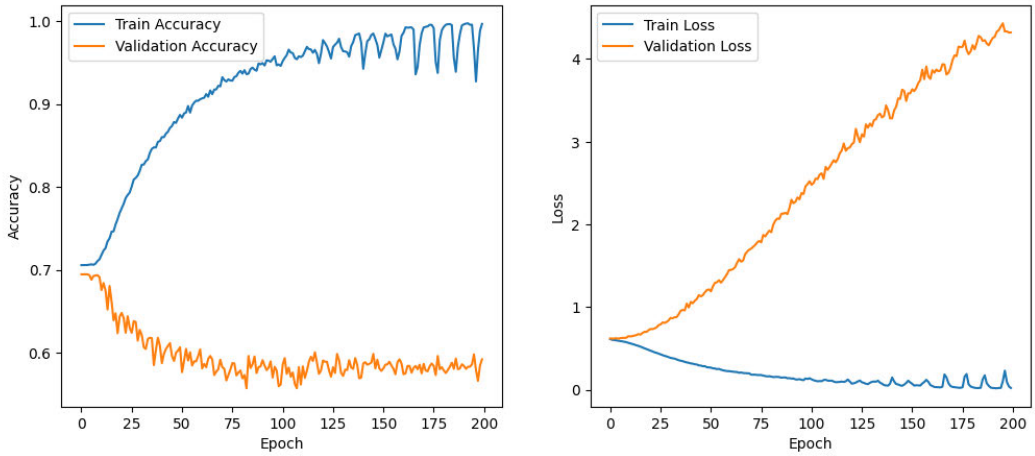


FIGURE 12. Training-validation accuracy and training-validation loss of the attention model.

the attention layer filters and emphasizes the relevant features. The attention mechanism assigns different relevance to various segments of the input sequences, enabling the model to concentrate on where it matters most. This function significantly enhances the efficiency of the model in understanding the dependencies and intricate relationships within the data.

The attention output is the flattened layer, which transforms the attended features into a 1-D vector, before passing through a full connection layer with 16 neurons for further feature abstraction. The last fully connected layer is a softmax activated layer that provides the probabilities of the classes assigned to a given problem as a binary classification issue. The model was then trained using the Adam optimizer and categorical cross-entropy loss for 200 epochs with a batch size of 32. This configuration assists in learning and guarantees generalization through validation monitoring. The strength of the attention mechanism as a way to concentrate on salient features makes this model a promising approach to modelling patterns in educational data.

Table 6 shows the assessment indicators for the attention-based model applied to the classification of student performance results. Class 0 (for students who were likely to fail) obtained a precision, recall, and F1-score of 0.33, thus implying high operation with nearly similar sensitivity and precision. When it comes to the classification of students among Class 1 (students predicted to pass), it has a relatively higher accuracy with a precision of 0.70, recall of 0.71 and F1-Score of 0.71 is good enough to conclude that the majority of the class is identified.

The accuracy of the model was 0.59, indicating moderate results in making predictions for a dataset with 2000 samples. The 0.51 of Macro Avg Precision, Avg Recall, and Avg F1-Score indicate that the accuracy of the model is evenly split between both classes; the weight average score of 0.59 also suggests a balanced model output. However, the classification of Class 1 appears reasonable based on the model, whereas there is scope for improving the classification of Class 0. Figure 12 shows the training and validation accuracies and loss of the CNN model.

TABLE 7. LSTM model evaluation metrics.

	Precision	Recall	F1-Score	Support
Class 0	0.30	0.28	0.29	611
Class 1	0.69	0.72	0.70	1389
Accuracy		0.58 (2000 samples)		
Macro Avg	0.50	0.50	0.50	2000
Weighted Avg	0.57	0.58	0.58	2000

H. LONG SHORT-TERM MEMORY PERFORMANCE AND PREDICTIONS

The network architecture had one LSTM layer containing 128 hidden units and one fully connected dense layer that employed a Softmax function for classification. Because the dataset had sequences of different lengths, padding was used such that all inputs to the model were of a standard size.

The optimization process used Adam with a learning rate of 0.001, because it offered a reasonable compromise between the speed of convergence and generalization. The model was set up with a multiclass categorical cross-entropy loss function for the same educational recommendations for different classes of students. To avoid overfitting the model, dropout regularization (0.2) was used after the LSTM layer. The training took 200 epochs, and an early stopping strategy supervised the validation accuracy. Training was stopped if, for five consecutive epochs, the model's validation accuracy did not improve, to avoid overfitting and computation waste. The memory and training stability were optimized with a batch size of 64.

The long short-term memory (LSTM) model was developed to consider dependencies in the sequence of data concerning student performance. The proposed model started with an LSTM network of 64 units, and its main function was to read the sequence and remember the crucial steps. This layer allows for deeper temporal feature extraction because it

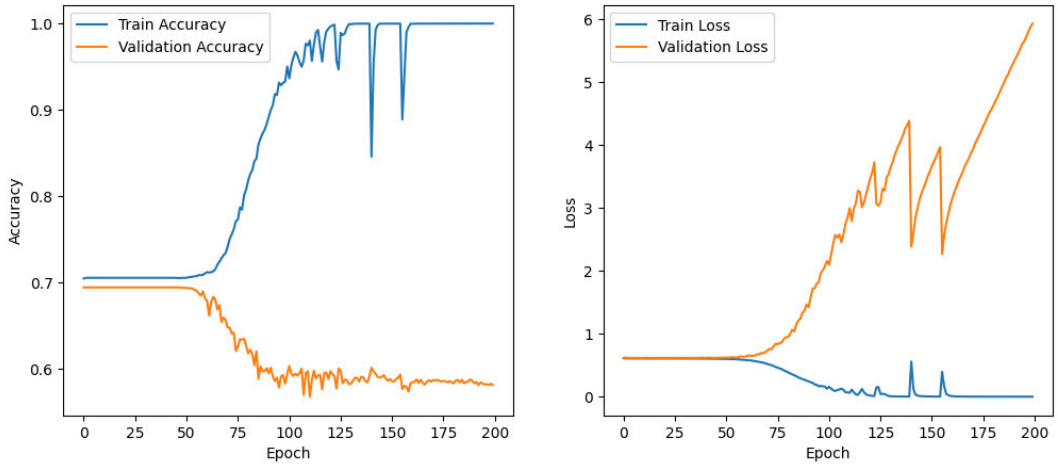


FIGURE 13. Training-validation accuracy and training-validation loss of the LSTM model.

is set to return sequences using return_sequences = true. The second LSTM layer had 32 units, took the packed sequential data in as the input, and returned a packed sequence as the output. This increases the ability of the stacked LSTM layers to capture multilayered temporal information in educational datasets, which may have different structures in different layers. The LSTM model integrates multiple gates and state mechanisms, which are mathematically formulated as follows.

The LSTM network processes sequential input data by using several key operations.

1) INPUT GATE

This gate controls the amount of new information added to the cell state:

$$\mathbf{i}_t = \sigma\left(\mathbf{W}_{ix}\mathbf{x}_t + \mathbf{W}_{ih}\mathbf{h}_{t-1} + \mathbf{b}_i\right)$$

2) FORGET GATE

The forget gate determines the extent of retention of past information:

$$\mathbf{f}_t = \sigma\left(\mathbf{W}_{fx}\mathbf{x}_t + \mathbf{W}_{fh}\mathbf{h}_{t-1} + \mathbf{b}_f\right)$$

3) CANDIDATE CELL STATE

A candidate state is computed to introduce new potential updates:

$$\tilde{\mathbf{c}}_t = \tanh\left(\mathbf{W}_{cx}\mathbf{x}_t + \mathbf{W}_{ch}\mathbf{h}_{t-1} + \mathbf{b}_c\right)$$

4) CELL STATE UPDATE

The current cell state integrates the input gate and forget gate information:

$$\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{c}}_t$$

5) OUTPUT GATE

The output gate filters the updated state to produce the output:

$$\mathbf{o}_t = \sigma\left(\mathbf{W}_{ox}\mathbf{x}_t + \mathbf{W}_{oh}\mathbf{h}_{t-1} + \mathbf{b}_o\right)$$

6) HIDDEN STATE UPDATE

The hidden state is computed as:

$$\mathbf{h}_t = \mathbf{o}_t \odot \tanh\left(\mathbf{c}_t\right)$$

I. DENSE AND OUTPUT LAYER TRANSFORMATION

After passing through the stacked LSTM layers, a dense layer with 16 neurons was applied to a nonlinear ReLU transformation as follows:

$$\mathbf{z} = \mathrm{ReLU}\left(\mathbf{W}_d\mathbf{h}_t + \mathbf{b}_d\right)$$

Finally, the Softmax activation function outputs the predicted probabilities.

$$\hat{\mathbf{y}} = \mathrm{softmax}\left(\mathbf{W}_o\mathbf{z} + \mathbf{b}_o\right)$$

The suggested layout was composed of profuse long short-term memory (LSTM) layers. Immediately thereafter, there was a single dense layer with 16 neurons and ReLU activation, which was responsible for performing nonlinear transformations and additional abstraction of the learned features. The output layer utilizes activation softmax with two neurons, which is used to predict the student's performance on Pass or Fail. The model was trained using the Adam optimizer in an adaptive manner and categorical cross-entropy loss over 200 epochs with a batch size of 32. The input data are transformed to match the format relevant to the LSTM processing input data type. Finally, the model was tested on a different dataset to make general comparisons for appropriate evaluation of the model patterns across increased time intervals in the dataset.

Table 7 lists the evaluation criteria of the LSTM model for recognizing student outcomes. For Class 0 (students predicted not to pass), we found a low precision of 0.30 and a low recall of 0.28, resulting in a very low F1-score of 0.29, which appears to be insensitive to this class, and a high percentage is misclassified as negative. However, in Class 1, the students who were predicted to pass showed a higher performance, with a precision of 0.69, recall of 0.72, and

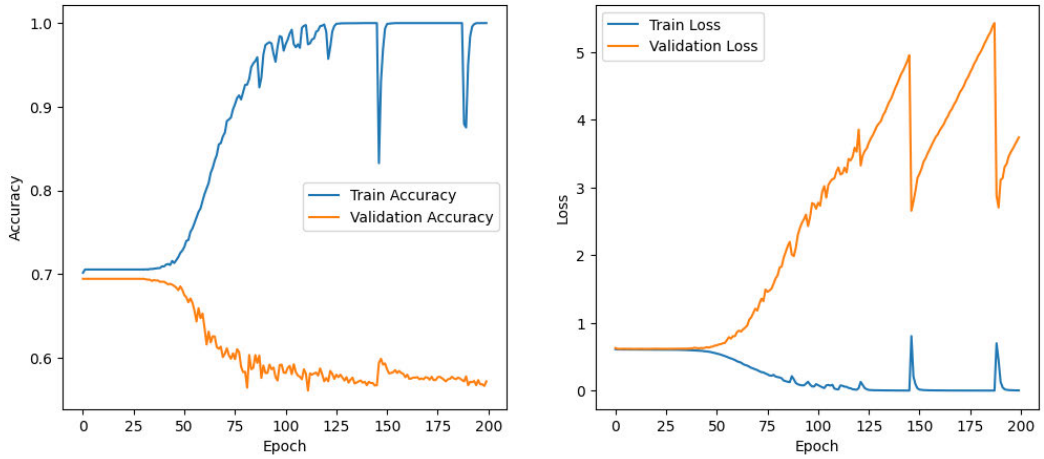


FIGURE 14. Training-validation accuracy and training-validation loss of the GRU model.

TABLE 8. GRU model evaluation metrics.

Metric	Precision	Recall	F1-Score	Support
Class 0	0.28	0.27	0.27	611
Class 1	0.69	0.71	0.70	1389
Accuracy		0.57		
Macro Avg	0.49	0.49	0.49	2000
Weighted Avg	0.56	0.57	0.57	2000

F1-score of 0.70, which indicates a better performance of the model in predicting the majority class.

The overall accuracy was 0.58; therefore, it exhibited a moderate performance when working with an overall dataset of 2000 samples. The macro average of each measure was 0.50, meaning that the model could classify instances of each class with fairly good accuracy, whereas the weighted average displayed the model's preference for the larger class Class 1. From these results, it is understood that although there is a fair understanding of temporal dependencies using LSTM models, there is a need for further upgrades in classification in general, and Class 0 in particular. Figure 13 shows the training and validation accuracies and loss of the LSTM model.

J. GATED RECURRENT UNIT (GRU)

The model architecture had one GRU layer with 128 hidden units, followed by a dense layer with Softmax for classification. Because the GRU of the network had a fixed-length requirement for sequences, zero padding was used to make all input sequences uniform for the GRU network. The model was trained with Adam, using a learning rate of 0.001 for fast convergence and stability. A categorical cross-entropy loss function was used to address the multiclass classification problems. To avoid overfitting, dropout (0.2) was used after the GRU layer. The model was trained for 200 epochs with early stopping based on validation accuracy. Training was stopped if the validation accuracy did not increase over five epochs, to reduce unnecessary calculations. A batch size of 32 was chosen to optimally use memory while maintaining stable training.

The GRU model applied in this study follows a sequential structure for input sequence classification and employs two Gated Recurrent Unit (GRU) layers to capture temporal relationships. The model starts with a GRU layer of 64 units, whose input shape is determined by the input_shape input sequences. The return sequences are set to True so that this layer can produce the entire output sequence, and the next GRU layer continues to capture dependencies over the inputs. The second-layer GRU consisted of 32 units, from which we passed another fully connected layer of 16 units to introduce nonlinearity through the ReLU activation function. The last layer in the model is a dense layer with softmax activation and thus produces probabilities in two classes. This architecture is intended for use in binary classification problems in which sequences of data values are present. The mathematical operation of the GRU is as follows:

The GRU network processes sequential input data using the following operations.

1) UPDATE GATE

The update gate controls how much of the past information needs to be passed along to the future:

$$\mathbf{z}_t = \sigma\left(\mathbf{W}_{zx}\mathbf{x}_t + \mathbf{W}_{zh}\mathbf{h}_{t-1} + \mathbf{b}_z\right)$$

2) RESET GATE

The reset gate determines how much of the previous hidden state contributes to the current candidate state:

$$\mathbf{r}_t = \sigma\left(\mathbf{W}_{rx}\mathbf{x}_t + \mathbf{W}_{rh}\mathbf{h}_{t-1} + \mathbf{b}_r\right)$$

3) CANDIDATE HIDDEN STATE

The candidate hidden state computes a potential update based on the reset gate and current input:

$$\tilde{\mathbf{h}}_t = \tanh\left(\mathbf{W}_{hx}\mathbf{x}_t + \mathbf{r}_t \odot \left(\mathbf{W}_{hh}\mathbf{h}_{t-1}\right) + \mathbf{b}_h\right)$$

4) HIDDEN STATE UPDATE

The final hidden state is a linear interpolation between the previous hidden state and the candidate state, weighted by the update gate:

$$\mathbf{h}_t = \mathbf{z}_t \odot \tilde{\mathbf{h}}_t + (1 - \mathbf{z}_t) \odot \mathbf{h}_{t-1}$$

K. DENSE AND OUTPUT LAYERS

After the GRU layers, a dense layer and output layer process the data for classification.

1) DENSE LAYER TRANSFORMATION

A dense layer with 16 neurons performs a nonlinear ReLU transformation:

$$\mathbf{z} = \text{ReLU}\left(\mathbf{W}_d \mathbf{h}_t + \mathbf{b}_d\right)$$

2) OUTPUT LAYER PREDICTION

A softmax activation function outputs the probabilities.

$$\hat{\mathbf{y}} = \text{softmax}\left(\mathbf{W}_o \mathbf{z} + \mathbf{b}_o\right)$$

During training, the model was compiled using the Adam optimizer, because it automatically adjusted the learning rate during training to achieve more rapid convergence. This is a multiclass classification problem; thus, categorical cross-entropy was used as the loss function and accuracy was selected as the primary performance measure. During training, the changes were processed for more than 200 epochs using a batch size of 32 while using validation data for the general performance of the model. Recording the history_gru variable makes it easier to check the training as well as the validation accuracy and loss, which is important when determining the extent to which the model may overfit. This study focuses on learning architectures that include recurrent and convolutional models, as well as training configurations to achieve high levels of accuracy when classifying sequential data.

Table 8 presents different evaluation parameters for the binary classification task related to the GRU model. The performance of the model for each class (i.e., classes 0 and 1) was evaluated in terms of the precision, recall, and F1 score. Owing to this imbalance, better results were observed for Class 1, with a precision of 0.69, recall of 0.71, and F1 score of 0.70, whereas Class 0 consisted of only 611 instances. The overall accuracy of the model was 57 per cent. The final result, which is the macro-average of the Precision, Recall, and F1-Score, was 0.46, which is the average of both classes without considering their size. However, the weighted model (0.56 precision, 57 recall, and 0.57 F1-score) corrects for the class distribution, indicating that the model is more accurate when predicting a larger class category. Figure 14 shows the training and validation accuracies and the loss of the GRU model.

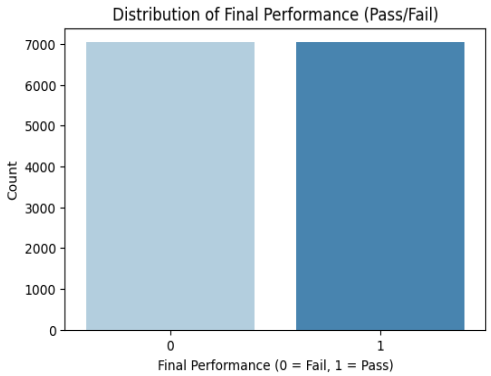


FIGURE 15. Final performance feature class distribution after data balancing.

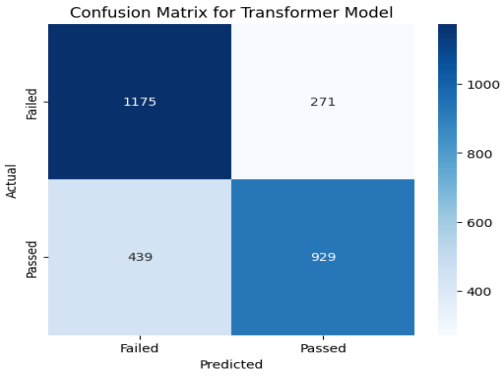


FIGURE 16. Confusion matrix for transformer model after data balancing.

IV. UTILIZING TRANSFORMERS FOR GENERATING ADAPTIVE LEARNING PATHS

To further improve the performance of the transformer model in generating adaptive learning paths, we performed the following steps.

- Dataset Balancing
- Training and evaluating transformer model on balanced dataset.

A. DATASET BALANCING

One of the primary issues highlighted in the dataset is the lack of proportionate balance in the final_performance variable, which is referred to as the dependent feature. First, there was a class imbalance issue within the dataset, where one class was represented excessively compared to the other. Such an imbalance may result in biased model performance, in which case the classifier will favor the majority class and reduce its ability to accurately predict cases in the minority class. To resolve this problem and improve model generalization, the author applied the synthetic minority oversampling technique (SMOTE).

Unlike other simple oversampling techniques, SMOTE synthesizes new examples in the minority class instead of simply copying existing examples. This method uses

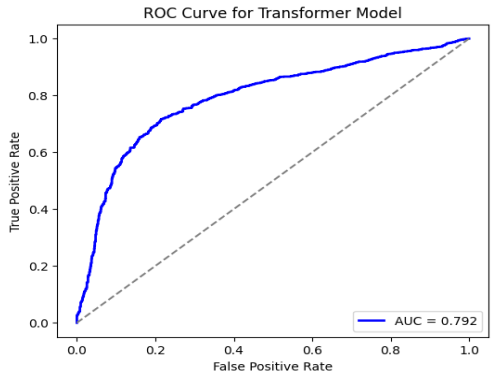


FIGURE 17. AUC-ROC for the transformer model after data balancing.

k-nearest neighbors (KNN) to create new candidates for the transformed target attribute within the domain of known examples. In this study, SMOTE was used to enhance the balance of the final_performance feature by oversampling records of both class labels (0 and 1) to 7000 each. Thus, the dataset was able to achieve a balanced class distribution without losing the original structure and statistical properties.

After applying SMOTE, the dataset was balanced, ensuring that there was no bias with respect to the training of the model. This measure assisted with performance classification, mitigating overfitting to the majority class and ensuring that the prediction model had good generalization capability for new data. The balanced dataset served as an excellent starting point for model training with machine-learning algorithms, which produced more accurate and unbiased results. Figure 15 illustrates the class distribution of the final_performance features for the pass and fail students.

B. TRAINING AND EVALUATING TRANSFORMER MODEL ON BALANCED DATASET

Once the data balancing strategies were applied, the transformer model was trained and evaluated to ascertain whether there was any improvement in generalization and robustness. The dataset was prepared by normalizing the numerical features and transforming categorical labels into one-hot vectors. I designed a transformer model with an encoder block that contained multihead attention, feed-forward layers, and layer normalization. The model was trained for 200 epochs with categorical cross-entropy loss using the Adam optimizer. The test set was evaluated for performance using accuracy, precision, recall, F1 score, and AUC-ROC and evaluated using a confusion matrix. Figure 16 and 17 shows the confusion matrix and AUC-ROC curve for the transformer model. Table 9 presents the classification report of the transformer model.

C. UTILIZING TRANSFORMER FOR GENERATING ADAPTIVE LEARNING PATHS

Figure 18 shows the complete workflow of the adaptive learning paths generated by the transformer model. The parameters specified were task difficulty, engagement rates, past success, and amount of time spent by students on tasks. These parameters were intended to reflect the heterogeneity in students' learning patterns with an emphasis on meeting the needs of specific disabilities. Such disabilities include Dyslexia, Attention deficit hyperactivity disorder, Autism Spectrum Disorder, Visual impairment, Hearing Impairment, and Intellectual disabilities.

TABLE 9. Classification report for the transformer model showing precision, recall, and F1-score for each class. The model achieved an overall accuracy of 75%, indicating balanced performance across both classes.

Class	Precision	Recall	F1-Score	Support
0 (Failed)	0.73	0.81	0.77	1446
1 (Passed)	0.77	0.68	0.72	1368
Accuracy		0.75		2814
Macro Avg	0.75	0.75	0.75	2814
Weighted Avg	0.75	0.75	0.75	2814

Mathematical Foundation for Adaptive Learning Path Generation using Transformers: We employed a transformer-based deep-learning architecture to design adaptive learning paths tailored specifically for visually impaired students. The following sections describe key equations and their significance in this context.

1) INPUT REPRESENTATION AND ENCODING

The input to the transformer model includes structured and unstructured data such as learner profiles, task descriptions, and resource metadata (e.g., difficulty level and accessibility features). This input is tokenized and embedded into a continuous vector space.

Input Embedding:

$$\mathbf{X} = \text{Embed}(\text{Tokenize}(\text{Input})) \tag{1}$$

where:

- Input represents the data elements fed into the model.
- Tokenize$(\cdot)$ splits textual and categorical data into discrete tokens.
- Embed$(\cdot)$ maps tokens into dense vector representations $\mathbf{X} \in \mathbb{R}^{n \times d}$, with n as the sequence length and d as the embedding dimension.

Positional Encoding:

$$\text{PE}_{(i,2k)} = \sin\left(\frac{i}{10000^{2k/d}}\right),$$
$$\text{PE}_{(i,2k+1)} = \cos\left(\frac{i}{10000^{2k/d}}\right) \tag{2}$$

where i is the position index and k is the embedding dimension index. These encodings were added to $\mathbf{X}$, allowing the model to learn the positional dependencies.

2) SELF-ATTENTION MECHANISM

Self-attention enables the model to capture the relationships between the input tokens, which are critical for generating paths that prioritize accessibility and content suitability.

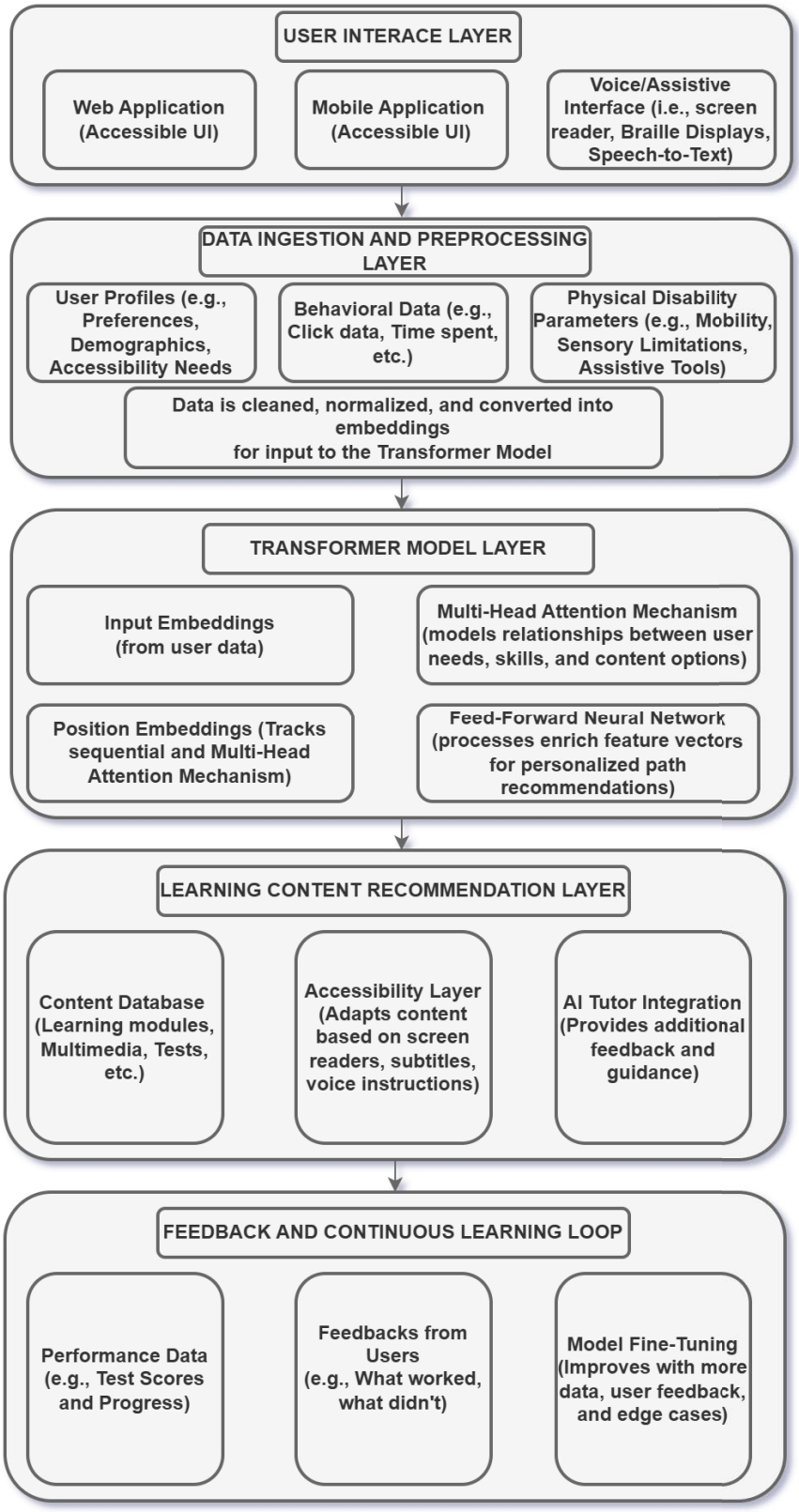


FIGURE 18. Transformer architecture diagram: adaptive learning path generator.

Scaled Dot-Product Attention:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d_k}}\right)\mathbf{V} \tag{3}$$

where:

- $\mathbf{Q} = \mathbf{X}\mathbf{W}^Q$, $\mathbf{K} = \mathbf{X}\mathbf{W}^K$, $\mathbf{V} = \mathbf{X}\mathbf{W}^V$,
- $\mathbf{W}^Q, \mathbf{W}^K, \mathbf{W}^V \in \mathbb{R}^{d \times d_k}$ are learnable projection matrices.

Attention Weight:

$$\alpha_{ij} = \frac{\exp\left(\frac{\mathbf{q}_i \cdot \mathbf{k}_j}{\sqrt{d_k}}\right)}{\sum_{j=1}^{n} \exp\left(\frac{\mathbf{q}_i \cdot \mathbf{k}_j}{\sqrt{d_k}}\right)} \tag{4}$$

Here, α_{ij} represents the attention weight between token i and token j.

3) MULTI-HEAD ATTENTION

To learn multiple types of relationships, we used multihead attention, where attention is computed in parallel across multiple subspaces.

Multi-Head Attention:

$$\text{MultiHead}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Concat}(\text{head}_1, \ldots, \text{head}_h)\mathbf{W}^O \tag{5}$$

where:

$$\text{head}_i = \text{Attention}(\mathbf{Q}_i, \mathbf{K}_i, \mathbf{V}_i) \tag{6}$$

and $\mathbf{W}^O \in \mathbb{R}^{hd_k \times d}$ projects concatenate the outputs back to their original dimensions.

4) FEEDFORWARD NETWORK

Each token embedding is passed through a position-wise feedforward network to capture the nonlinear transformations.

Feedforward Network:

$$\text{FFN}(\mathbf{z}) = \sigma(\mathbf{z}\mathbf{W}_1 + \mathbf{b}_1)\mathbf{W}_2 + \mathbf{b}_2 \tag{7}$$

where:

- $\mathbf{W}_1 \in \mathbb{R}^{d \times d_{ff}}$, $\mathbf{W}_2 \in \mathbb{R}^{d_{ff} \times d}$,
- $\mathbf{b}_1, \mathbf{b}_2$ are biases,
- $\sigma(\cdot)$ is a non-linear activation function (e.g., ReLU).

5) LOSS FUNCTION FOR ADAPTIVE LEARNING

The learning path prediction model optimizes a weighted loss function, balancing the content relevance and accessibility.

Generalized Loss Function:

$$\mathcal{L} = \mathcal{L}_{\text{content}} + \lambda_1 \mathcal{L}_{\text{accessibility}} + \lambda_2 \mathcal{L}_{\text{user_feedback}} \tag{8}$$

where:

- $\mathcal{L}_{\text{content}}$: Cross-entropy loss for learning path prediction,
- $\mathcal{L}_{\text{accessibility}}$: Penalty for non-compliance with accessibility constraints,
- $\mathcal{L}_{\text{user_feedback}}$: Additional penalty based on learner feedback (e.g., satisfaction scores).

6) OUTPUT GENERATION

The model generates learning paths using an autoregressive decoder conditioned on encoded representations.

Next-Step Prediction:

$$P(y_t | y_{<t}, \mathbf{X}) = \text{softmax}(\mathbf{h}_t \mathbf{W}^O) \tag{9}$$

where:

- $\mathbf{h}_t$: Hidden state at time t,
- $\mathbf{W}^O$: Output projection matrix.

7) EVALUATION METRICS

We used weighted accuracy metrics that prioritized accessibility compliance to ensure alignment with the needs of the visually impaired learners.

Weighted Accuracy:

$$\text{Accuracy}_{\text{weighted}} = \frac{\sum_{i=1}^{N} w_i \cdot \mathbb{I}(\hat{y}_i = y_i)}{\sum_{i=1}^{N} w_i} \tag{10}$$

where:

- w_i: Weights for each learner or instance (e.g., based on impairment severity),
- $\mathbb{I}(\hat{y}_i = y_i)$: Indicator function for correct predictions.

8) INCORPORATING ACCESSIBILITY CONSTRAINTS

Adaptive learning paths for visually impaired students must adhere to accessibility standards in order to ensure that audio, tactile, and visual enhancements are embedded.

Accessibility Penalty:

$$\mathcal{L}_{\text{accessibility}} = \sum_{j=1}^{M} c_j \cdot \mathbb{I}(\text{constraint}_j \text{ violated}) \tag{11}$$

where:

- c_j: Weight of constraint j,
- constraint$_j$: Accessibility-specific rule (e.g., availability of audio descriptions).

For each student, three types of interactions with tasks were created using random data:

- On a task difficulty scale, the integer range was between 1 and 5.
- Engagement score was a scalar in the interval 0.5 – 1.0.
- Past performance was 'success' and 'failure' coded as '1' and '0' respectively.
- Time spent performing tasks ranged from 1 minute to 15 minutes.
- The response was binary and based on correct response completion.

The total responses were calculated and averaged as the active dependent variable of interest and the number of correct responses was averaged. After such an operation, the target variable comprises two classes: 0 or 1.

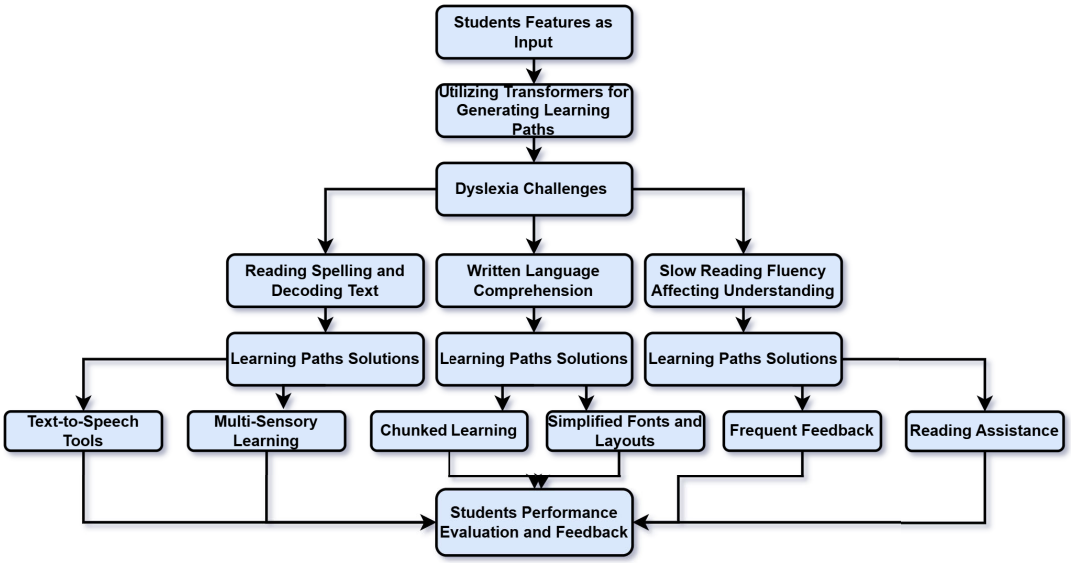


FIGURE 19. Adaptive learning paths for students with dyslexia learning disorder.

D. ADAPTIVE LEARNING PATH GENERATION

As part of the intervention, students identified as struggling underwent task performance evaluation based on set learning objectives. The suggested strategies include the following.

- Encouragement and practice for tasks classified as tough.
- Integrating multi-sensory content to improve engagement.
- Providing students with additional time to complete tasks.
- Frequent and purposeful evaluation targeting the change of strategies.

Figure 19 shows how the transformer model generates adaptive learning paths for students with dyslexia. Table 10 presents details of the adaptive learning paths generated by the transformer model. Similarly, Table 11 presents the adaptive learning paths generated for the students with ADHD.

Moreover, the transformer model was used to formulate adaptive learning paths for students with Autism Spectrum Disorder (ASD), Visual Impairment and Hearing Impairment. This entails training the transformer model with several specific student features, enabling it to understand the more intricate associations between the requirements of each individual and the most effective educational modifications. The approach can be outlined in two key steps: (1) training the transformer on special needs features and (2) implementation of the transformer in the development of adaptive learning paths.

The training data of the transformer model included a large array of data points, forming an understanding of the learning needs, difficulties, and preferences of students with disabilities. Some salient features include the following.

1) SENSORY SENSITIVITIES

which include determinants of students' attention and response to various sensory inputs, most especially for students with ASD who have been identified as requiring specific sensory learning environment characteristics.

2) COMMUNICATION ABILITIES

which consists of information on how a particular student communicates through sign language or gestures that are needed to develop learning tools for students with hearing or sight impairment or both.

3) VISUAL AND AUDITORY ACCESS

Information on the client's era of visual and hearing incapacity, the inclination that aids in determining the model's assistive device recommendation and alternate content version formats.

4) BEHAVIORAL AND EMOTIONAL REGULATION

Information on student behaviour and the need for emotional control often presents a challenge to students with ASD and need assistance in learning contexts.

The self-attention of the transformer model was used to capture the dependencies among these features, helping the model establish an intricate binding between individual student profiles and the most desirable types of support. For a single series of continuous supervised learning processes, the transformer model was taught regarding which educational strategies and instrumentation were connected to which set of tasks. More practically, the model showed that students suffering from great sensitivity to stimuli benefit from well-ordered environments, visual timetables, and sensory breaks, whereas students with sight impairment vision needed tactile materials, braille displays, and corresponding orientation

TABLE 10. Adaptive learning path for students with Dyslexia.

Dyslexia	Details
Challenges:	• Reading, spelling, and decoding text is surprisingly difficult. • A considerable struggle with the written language and understanding what has been written. • Reading fluency is slow which in turn hinders comprehension.
Learning Paths:	**Text-to-Speech Tools:** • Use tools such as Read&Write or Kurzweil 3000 which can read text out loud to help decode written language. • The application of emphasizing the words when reading so that the students can reinforce word recognition. **Multi-Sensory Learning:** • A combination of visual, auditory, and kinesthetic modalities such as sand writing letters, tracing letters, and flashcards with pictures. • Using applications such as Nessy Learning Program to have an interactive multi-sensory practice. **Chunked Learning** • Content should be chunked into smaller sections to enhance understanding. • Text should be supplemented with visual and step-by-step guides. **Simplified Fonts and Layout:** • Use dyslexia-friendly fonts like OpenDyslexic, and contrasting types of layout. • Don't cram too much text; use spacing and bullet points where appropriate. **Frequent Feedback:** • Students should perform formative assessments and be given feedback to help improve their confidence. • Students should receive constructive feedback after making spelling or word usage errors. **Reading Assistance:** • Students should be paired with their classmates or AI adaptive reading tools which change difficulty according to progress.

training. The adaptive learning paths generated for ASD, visual impairment, and hearing impairment are shown in figure 20, 21 and 22, respectively.

E. USING EXPLAINABLE ARTIFICIAL INTELLIGENCE TO UNDERSTAND STUDENTS WITH HEARING IMPAIRMENT

We adopted a multiphase strategy to explore the academic challenges faced by hearing-impaired students by integrating the principles of supervised machine learning and eXplainable Artificial Intelligence (XAI) methods. The methodology entails the process of data cleaning, model development, and use of different XAI approaches to retrieve informative explanations for the model's outcomes.

The dataset used in this study included various factors related to demographic characteristics, educational background, and other types of assistance offered to children with hearing impairment. The main features were age, sex, financially active background, type of school attended (special or regular), number of students in classes, number of interpreters available, hours of therapy attended, use of AT, level of hearing loss, presence of other disabilities, parents' participation, and achievement levels (the results of the tests). For phase three, categorical variables were dummy-coded, while continuous variables were z-rationed to assist in the comparison of the parameters.

The main predictive model adopted was a Random Forest Regressor because of its efficient performance in mixed-type data, its characteristics of not easily overfitting, and its ability to provide feature-importance scores. To obtain trustworthy assessments, the dataset was divided into training and testing subsets in an 80:20 ratio. The performance of the model was evaluated using the mean squared error. After the model

TABLE 11. ADHD challenges and adaptive learning paths generated by transformer.

Category	Description
Challenges	• Persistent attention concerns along with time organization. • Uncontrollable agitated tendencies. • Has difficulties switching between tasks and concentrating on something.
Learning Path	**Gamed learning:** • Use sites like Kahoot or Classcraft to harness participation and focus. • Creative ideas like levels earned or badges can be encouraged to complete tasks.
	Short-one focused: • Teach in lessons with shorter duration breaks (10 – 15 minutes) in between. • Use timers to create time pressure on participants.
	Visual Schedule: • Daily visual with start positions and task end should be available for the participants or specific functions. • Icons of varying priority or colours can be used as visual aids.
	Channels Lessons: • Reduce commonly used channel lessons. • Practice visual and auditory confinement using noise-cancelling headsets or screened working stations.
	Practice behavioral reinforcements: • During task performance or improvement in focus, a token system or points/credit is expected.
	Mindfulness Exercises: • Include 6 practices: Integrating in summer activities and daily life practices. • Encourage the use of apps like Calm or Headspace for practising attention-focused exercises.

performed well in terms of prediction accuracy, the procedure was further explored using artificial intelligence techniques to determine how different characteristic variables and their combinations affected the outcome.

To examine the relative importance of the features for the predictions made by the Random Forest model, SHapley Additive exPlanations (SHAP) was used. The SHAP is a game-theoretic approach that works within a single coherent framework and provides global and local explanations of how each feature affects the output of the model. This was performed using a summary plot, which provides an aggregate of the total feature contributions to predict the output, as shown in figure 23. Furthermore, the figures of the model's variable importance scores (parental involvement) were correlated and explained by the degree of each intelligence goal achieved, with the predicted feature value as the dependent variable, as shown in figure 24. Finally, the feature importance scores embedded in the Random Forest model were collected and plotted, as shown in Figure 25, allowing the audience to compare the dependency of any given feature. With the combined approach of SHAP and the visualization of the features' importance, the understanding

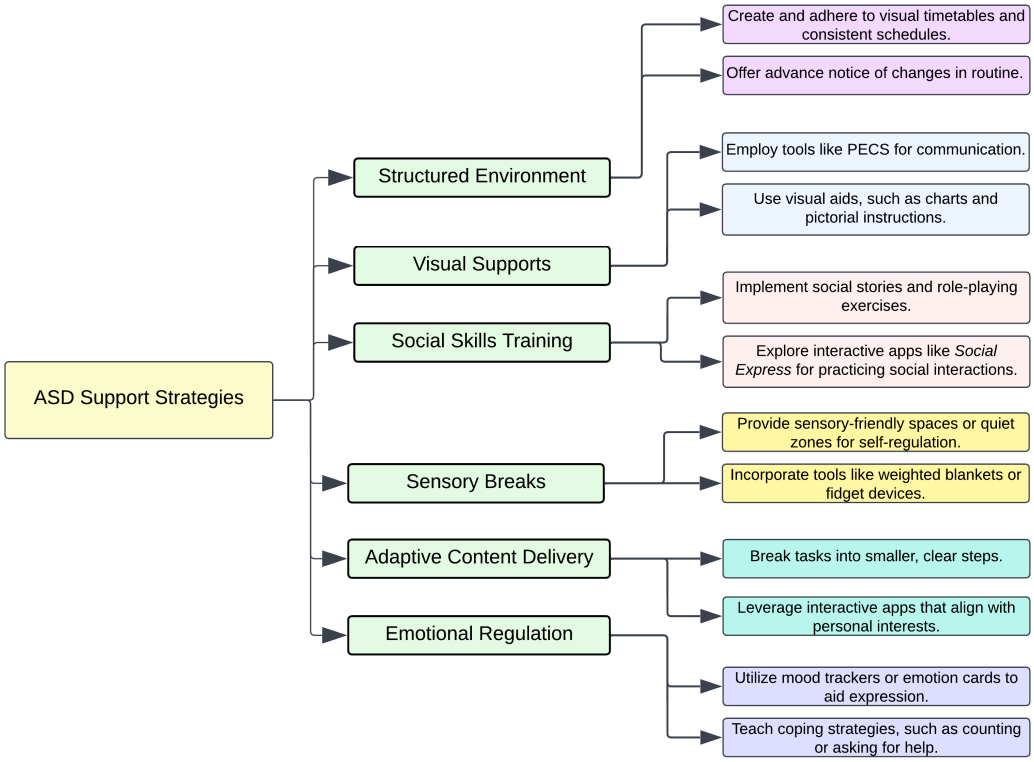


FIGURE 20. Adaptive learning paths generated by transformers for students with ASD.

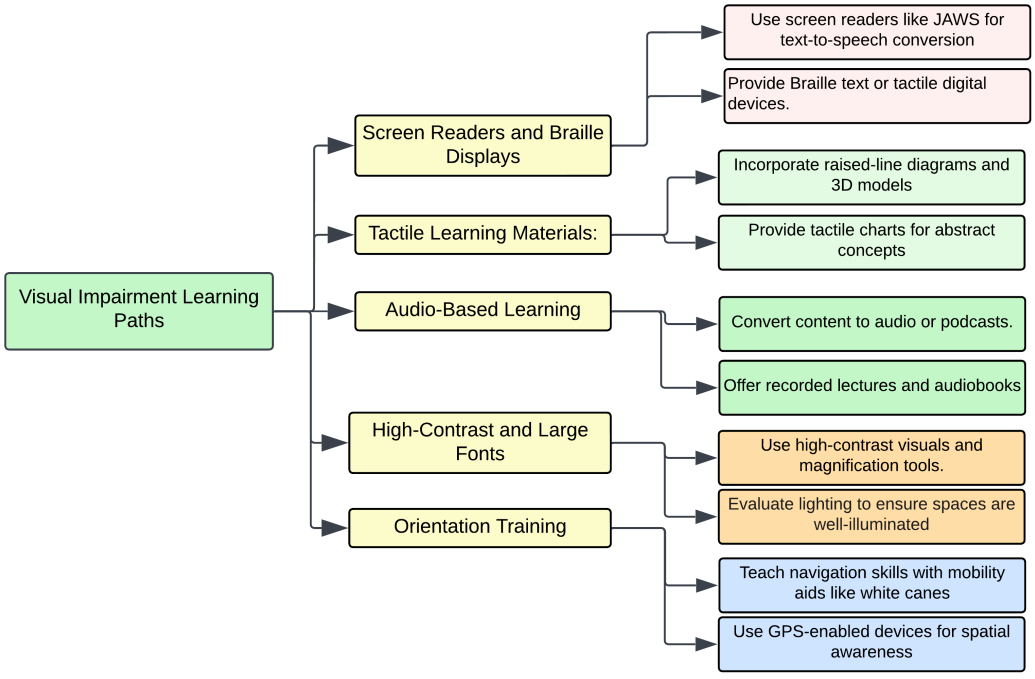


FIGURE 21. Adaptive learning paths generated by transformers for visually impaired students.

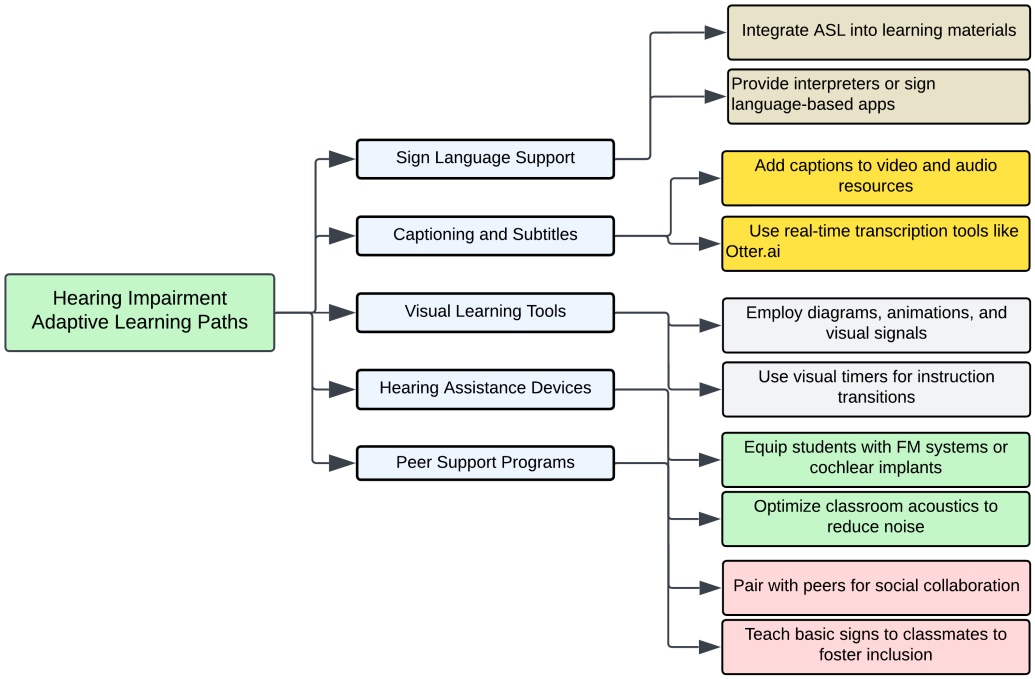


FIGURE 22. Adaptive learning paths generated by transformers for hearing impaired students.

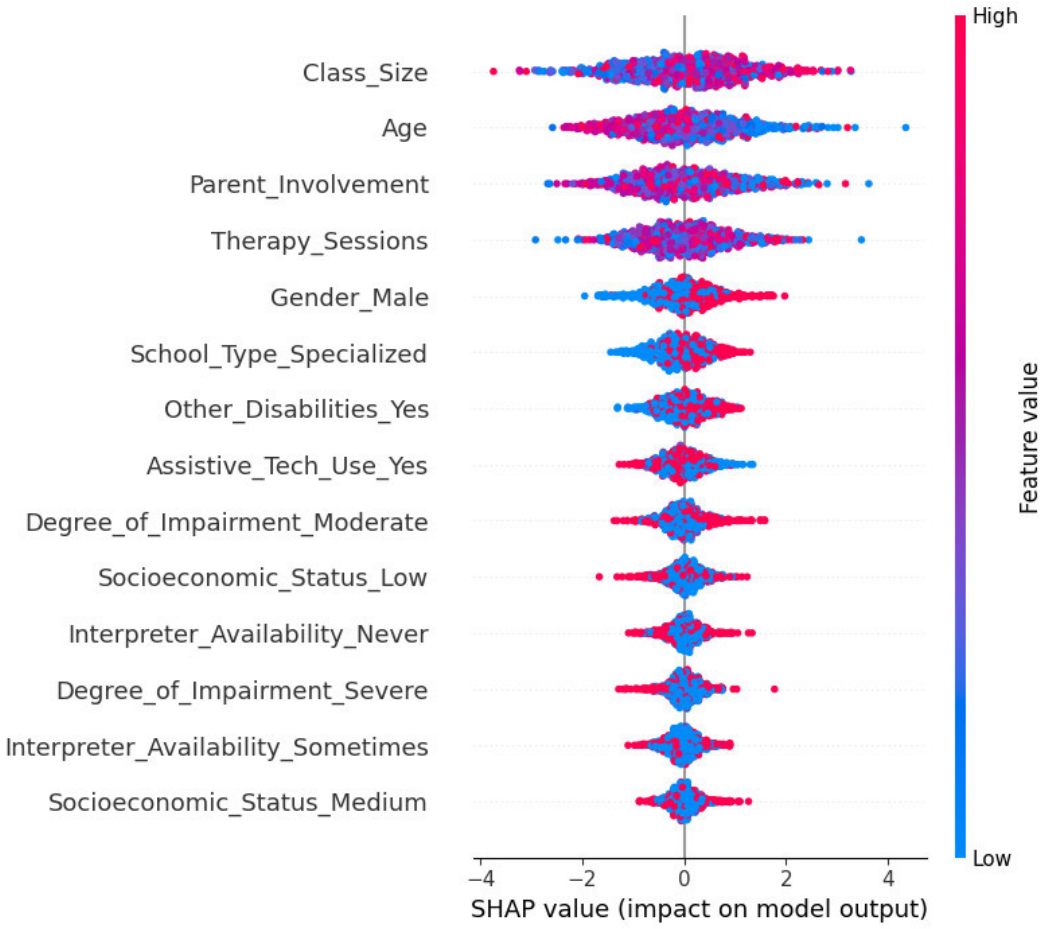


FIGURE 23. SHAP values impact on the model output.

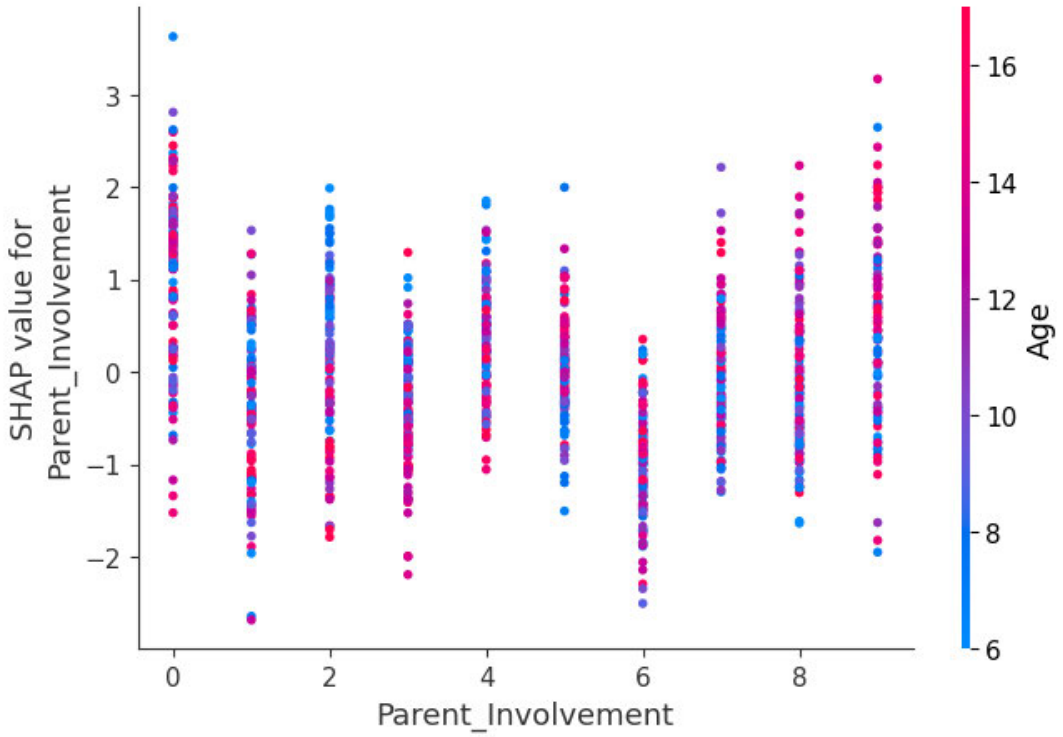


FIGURE 24. SHAP values for the parental involvement.

of the reasoning of the model is enhanced and transformed into specific usable knowledge.

To add interpretability to the Random Forest model, we applied the local interpretable model-agnostic explanation (LIME) framework. LIME's purpose of LIME is to interpret the prediction of a single instance given a complex model by fitting a local surrogate model, which is much simpler.

A special tabular explainer was prepared to interpret the tabular dataset along with the trained model to explain the actual estimate in the context of regression. The variables most responsible for individual observations in the data were elicited using the explainer and how they shaped the predicted outcome. For example, when the target for a single student's prediction was trained, ten predictors were found to be responsible for predicting the actual performance score, as shown in figure 26. This perspective is beneficial for comprehending more reasons for the model's recommendation in particular cases within a recommendation, which in turn makes it applicable for decision making.

Partial Dependence Plots (PDPs) were created to establish the individual infectiousness and impact of selected attributes, Parent Involvement, and Frequency and Duration of Therapy Sessions on the expected performance outcomes of students with hearing loss, as presented in Figure 27. Such plots present the strength of the relationship between the model predictions and each selected feature, while keeping all other features constant.

1) PARENT INVOLVEMENT

The graph then explains this phenomenon by showing that, as Parent Involvement increases, student performance is also expected to improve. The shifts in the curves caused by considerable variations in parental involvement level explain the model's predictive power for performance improvement.

2) THERAPY SESSIONS

Similarly, the number of therapy sessions attended demonstrated an active and dynamic effect on predicted outcome variables. The data showed that the therapy sessions had a nonlinear effect on student performance, as indicated by fluctuations in the plot.

To further investigate the variability in model predictions for the Parent Involvement feature, the partial dependence plots and the individual conditional expectation curves were combined in the PDP-ICE approach and plotted as shown in figure 28.

Partial Dependence Plot (PDP): The orange dashed line indicates the mean expected outcome for the response variable as predicted by the features across all data. This graph illustrates the overall impact of parental involvement on student success in a model's prediction, depicting the tendency of the model concerning how features change.

ICE Curves: The light-blue lines are measures of individual students, most of whom were students in the dataset used in the model. Parent Involvement affects the inclusion of high

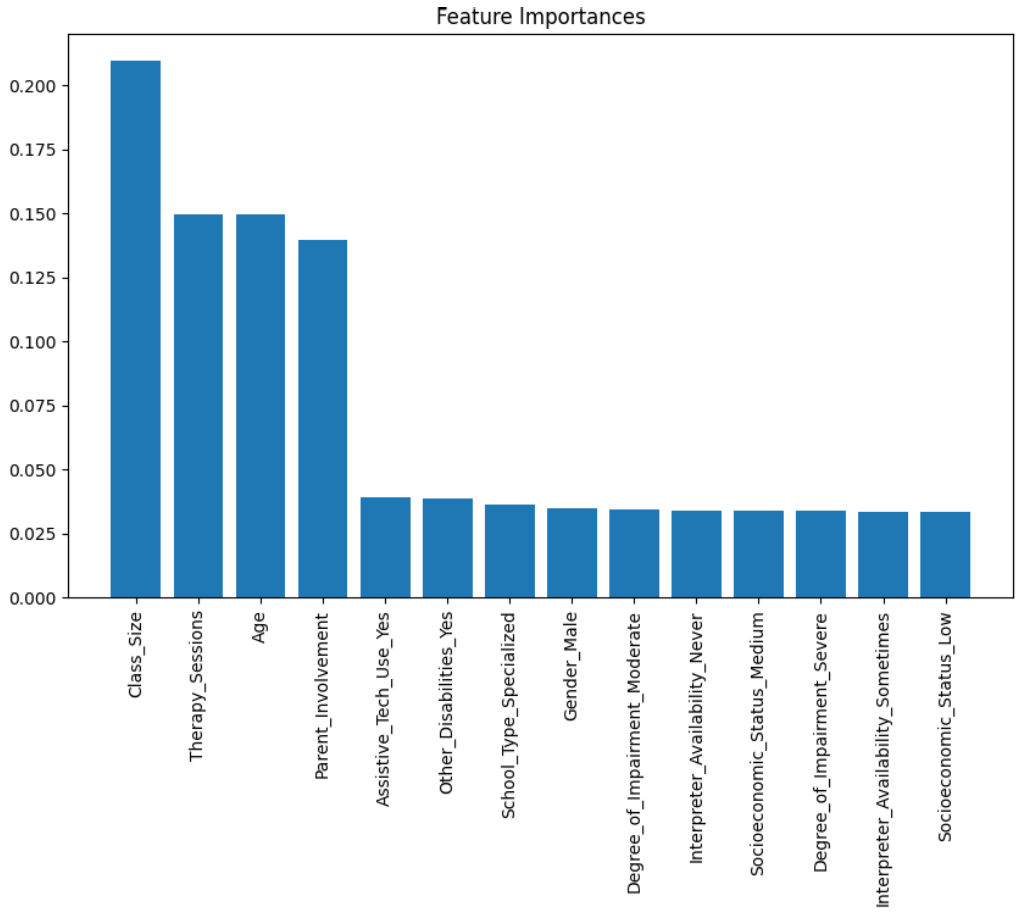


FIGURE 25. Relative feature importance influencing the final performance score of students with hearing impairment.

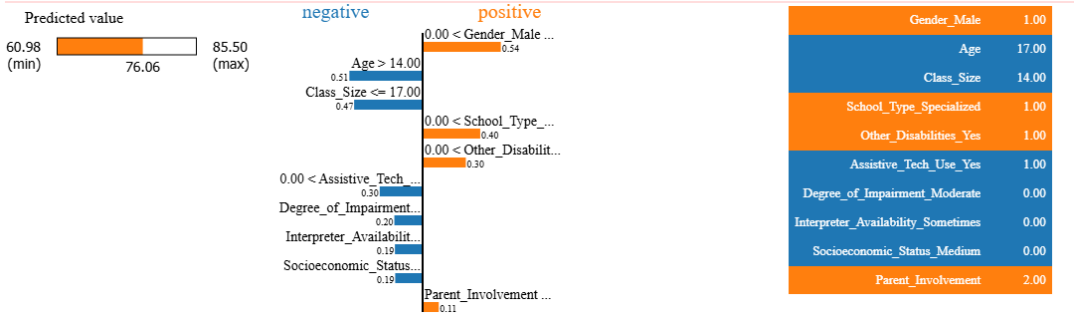


FIGURE 26. Local Interpretable Model-agnostic Explanations identified the most influential features for an individual data point, highlighting their contribution to the prediction.

student performance measures in the model. The uncertainty of these lines interprets the effect of the features on the students, indicating the diversity of the students in the dataset.

This two-pronged approach provided a deeper understanding of this issue. If all ICE explain such variability in behavioral assumptions, then the ICE would outshine the influence of the self-explanatory universality tendencies of the predominant component seen in focus. However, PDP usually explains the main tendency, while ICE captures the variability of the behavior and differences between people and allows us to draw conclusions about which students or subgroups students should belong to and which factor of Parent Involvement is most paramount, with others likely being the opposite. Such information is important for designing interventions that fit the individual or designed profiles of students.

Subsequently, the Decision Tree Regressor is applied as another global surrogate model to translate the outcome of a

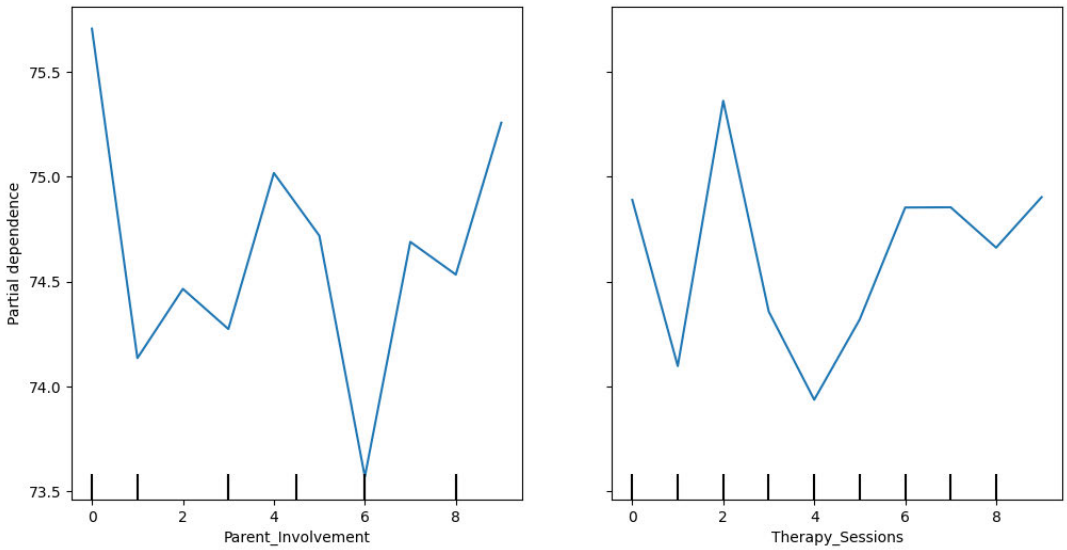


FIGURE 27. Partial Dependence Plots (PDPs) were used to analyze the effect of parent involvement and therapy sessions on the performance of students with hearing impairments.

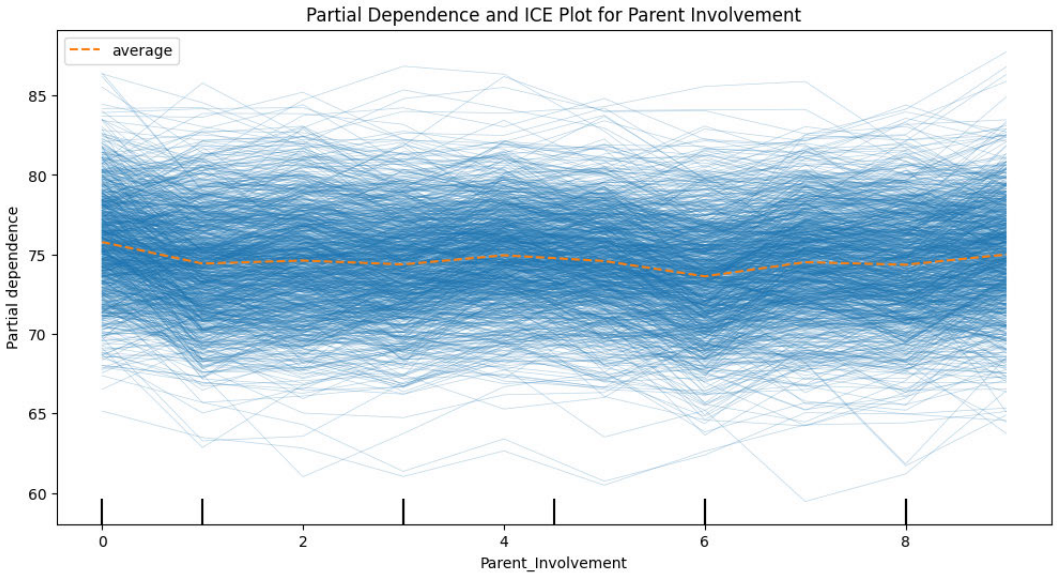


FIGURE 28. Partial dependence plots and the individual conditional expectation curves for understanding the variability in model predictions for the Parent Involvement feature.

more complex Random Forest model, as shown in figure 29. Thus, the surrogate model is constructed by modelling the same features as the test set (X_test) with target values as the Random Forest predictions instead of the ground true values. Stuck to a tree depth of three, the decision tree can be easily interpreted, where each split is a threshold-based choice of a specific feature (such as class size, age, and parent involvement). The visualization is the node for the squared error, which refers to the model fit, and the remaining nodes represent the number of samples that pass through the split and the average predicted value of the split. For instance, the initial split of the data separates the data based on class size, so subsequent splits focus on other aspects, such as age and parents' involvement in priority order. This visualization translates to the surrogate model's capability to mimic the abstract algorithm, while providing a glimpse of the cornerstone pattern and rules of the Random Forest.

V. COMPARISON OF MULTIMODAL AND PERSONALIZED ADAPTIVE LEARNING VS. SINGLE-MODE LEARNING APPROACHES

Based on recent studies, we analyzed single-mode learning systems alongside our personalized learning model, which is a proactive multimodal adaptive learning system. As the focus of our deep learning models (Neural Networks, GRU, LSTM, Attention-based Model, and Transformer),

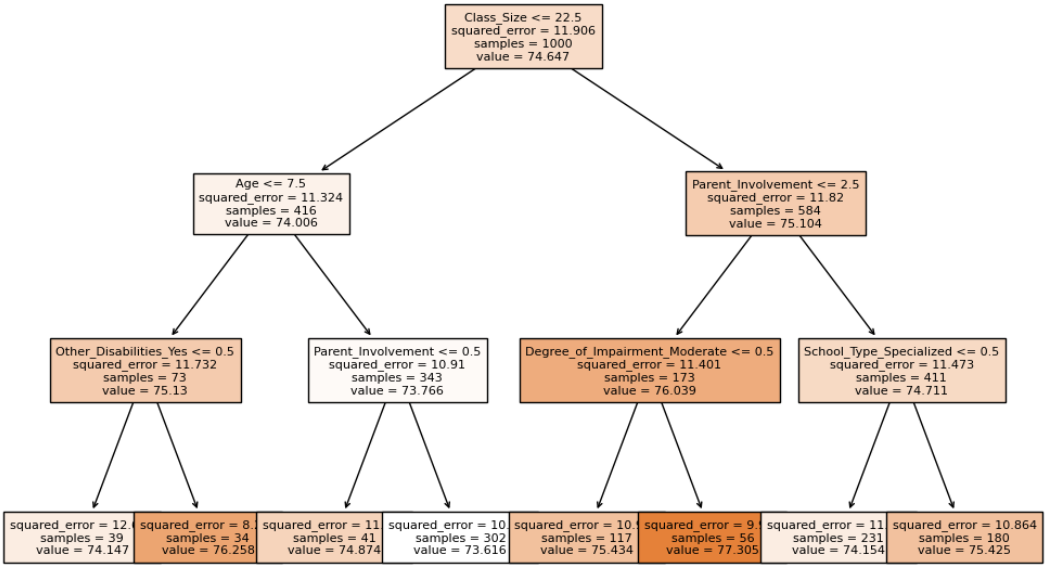


FIGURE 29. Decision Tree Regressor as a global surrogate model.

TABLE 12. Comparison of Multimodal Adaptive and Personalized Learning vs. Single-Mode Learning Approaches.

Feature	Our Study (Multimodal Transformer-Based Model)	Single-Mode Learning Approaches (References)
Learning Model	Transformer, LSTM, GRU, CNN, Attention-based Model	Single deep learning models (CNN, RNN, LSTM) [38]
Data Modalities	Text, speech, images, behavioral data (multimodal)	Text-only, speech-only [36]
Personalization	High – Uses multiple data inputs for personalized adaptive learning	Low – Limited to one data type (text, speech, or image)
Engagement & Retention	Increased engagement through multimodal input	Lower engagement due to lack of variety
Explainability (XAI)	Uses Explainable AI (XAI) to interpret student learning patterns	Limited interpretability in single-mode learning
Precision	**69.450% (Transformer)**	**75.3% (LSTM in single-mode learning)** [38]
Recall	**88.5% (Transformer)**	**1.00% (GRU in single-mode learning)**
F1-Score	**81.9711% (Transformer)**	**73.5% (CNN in single-mode learning)**
Overall Accuracy	**69.45% (Transformer-based Multimodal Learning)**	**78.2% (Single-mode learning models)**

we noticed that the performance of the transformer when generating adaptive learning paths met the criteria for precision, recall, F1-score, and accuracy.

A. KEY FINDINGS FROM LITERATURE ON SINGLE-MODE LEARNING

1) TEXT-BASED ADAPTIVE LEARNING SYSTEMS

These models operate through Natural Language Processing (NLP) techniques primarily and neglect to consider multimodal student engagement [35].

2) SPEECH-BASED LEARNING APPROACHES

Used in conducting verbal tests, they are devoid of any visual and behavioral elements which results in a considerably lower degree of individualization [36].

3) IMAGE-BASED LEARNING METHODS

Applied in recognizing handwriting or other forms of visual education, but are ineffective in teaching the concepts [37].

4) SINGLE DEEP LEARNING MODELS (LSTM, CNN, GRU)

These models are effective for sequential and spatial data separately, but in the presence of multimodal data, they fail. As with most single-modality models, the complex relationships between different data modalities are not captured [38]. Table 12 presents a comparison of multimodal adaptive and personalized learning with the single-model learning approaches.

VI. CONCLUSION, LIMITATIONS AND FUTURE WORK

The present research demonstrates the efficacy of deep learning principles within multimodal adaptive learning systems for students in the special education category. Several models were trained and assessed, including the baseline neural network model, Convolutional Neural Network (CNN), Attention Model, Long Short-Term Memory (LSTM), Gated Recurrent Units (GRU), and transformer models. We established that attention models and transformer architectures are superior to simpler neural architectures in terms of efficiency, generalization, and prediction capacity. This is primarily because of the transformer's self-attention mechanisms, which allow it to better integrate and process multiple domains of inputs consisting of academic performance metrics, behavioral attributes, and contextual aspects.

On the other hand, XAI techniques have opened the abstract of these models, allowing educators to gain visual insights into the features that matter most to their students. Such insights can help gain confidence in model recommendations and modify teaching strategies to suit the needs of students effectively.

This research illustrates the need to employ advanced neural network models, in this case transformers, to provide a more interactive and individualized learning experience for students with different learning needs. These findings herald the imminent arrival of sophisticated learning systems that are more precise and sensitive to the cognitive, behavioral, and affective attributes of the learner, thereby advancing the goals of any education.

Nonetheless, this study achieved its objectives, but had numerous shortcomings. First, although reasonably large, the training dataset does not necessarily represent real educational settings. Even the learning performance of the model has built-in data biases that compromise the generalization to other student groups and learning contexts. The dataset may be unrepresentative owing to a lack of demographic factors, such as socioeconomic status, cultural heritage, area, and even language, which may lead to bias in the model. These biases could hinder the generalization of the model across various student populations and learning contexts. For example, if the dataset mostly contains students from well-endowed schools in certain areas, the model will likely fail in less-fortunate or culturally different educational settings.

Further, the intricacy of the Transformer models means that a great deal of computational effort is required during their training and deployment, which could restrict scalability in schools or districts with a poor technical background. Moreover, transformer models require considerable computational power during training and deployment phases. This poses a problem for some schools or districts that do not have trained technical personnel or appropriate funding, thereby limiting the model's applicability in real-life educational settings. More powerful, resourceful, and emerging technologies will only complicate prevailing inequities. This also dampens the spirit of already burdened institutions. More research is needed on building lightweight models, and ensuring that the datasets are not biased towards certain regions or contexts is necessary to address these issues.

In conclusion, concerns about the interpretability of the model's decisions remain unaddressed regardless of the utility provided by XAI techniques, which is especially true for non-technical educators. The lack of availability of simple interpretation tools has broadened this gap. Finally, while the models showed very high predictive abilities, the extension of these findings into more practical and relevant actions in the form of teachers remains a task that requires the most attention.

Follow-up studies should attempt to find more representations by applying the models to students from diverse backgrounds, coupled with contexts that would make the models more fair. A view that assumes a wider educational system and more categories of special needs benefits from the development of extensive adaptive learning principles.

Follow-up studies should attempt to look for more representations by applying the models to students from diverse backgrounds, coupled with contexts that would make the models more fair. The development of more extensive adaptive learning principles would benefit from a view that assumes a wider educational system and more categories of special need.

Additionally, it would be interesting to look into ways to further extend existing real-time adaptive learning models so that they can react to changes in a student's learning trajectory within the shortest possible time. Combining reinforcement learning with transformer architectures may provide a means of adjusting learning paths to facilitate the creation of an even more responsive learning environment.

Apart from enhancing technical robustness, advancements in XAI are also needed to create more practical and interactive visualizations of data that non-experts such as educators can easily appreciate and act upon. AI scientists, education experts, and psychologists should work together to design these visualizations and ensure that they meet instructional requirements.

Finally, the scope of research should include factors that enhance the development of affordable and lightweight transformer models to enhance the availability of such systems within under-resourced educational contexts. Strategies such as model pruning, quantization, and distributed training would, in turn, reduce computational costs and increase accessibility for many more learners.

In summary, a focus on such cross-sectional collaboration, as well as on relevant technological progress, would be necessary to bring these adaptive learning systems to a wider audience, particularly students with special needs.

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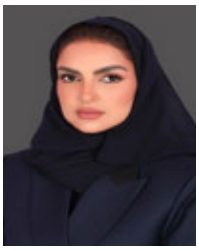
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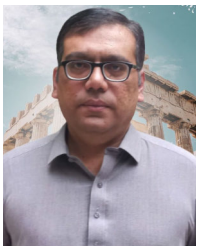


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